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The Brockmeier Model

How much risk to take at each age when stocks are priced to beat TIPS by two points a year, not six

Charlie Brockmeier · 36 min read · 19 charts, 14 tables

Prices as of November 9, 2026 · Companion: The Illiquidity Premium Changed Hands

Views are my own and not those of my employer. Nothing here is investment advice. Every figure comes from a public source cited at the end, or is labeled as my own arithmetic; none comes from my employer's data or products, or from any client.

SUMMARY

1. **The reward for risk is about a third of what the textbooks use.** On November 9, 2026 ten-year TIPS yielded 2.66% after inflation and thirty-year TIPS 3.08% [1]. Global stocks are priced to compound at about 4.61% a year (my arithmetic from index earnings and dividend yields [4-6, 9]). That is 1.95 points more than TIPS. From 1928 to 2025 U.S. stocks compounded 6.45 points a year faster than Treasury bills [12]. As arithmetic means, which is what the portfolio formula takes, the two premiums are 3.09 and 8.21 points.

2. **So the "60/40 investor" should hold a third to under a half in stocks once the paychecks stop.** The Merton share is the premium divided by risk aversion times variance [22]. The risk aversion that gives 55% in stocks on the historical averages gives 33.9% on today's prices. Measured against Treasury bonds instead of bills, the old 60% becomes 45%. The rest belongs in TIPS, not in nominal bonds.

3. **Age matters because of the savings you have not made yet.** Future savings are an asset, and for stable pay they behave like a bond. The model has a balanced saver all in stocks until 34, at 66% by 40 and 34% at 65. The largest target-date series holds 90% at 41 and about half at 65 [32]. Run on the 1928-2025 averages, the same model gives 99% at 40 and 55% at 65: a target-date fund is this model with stale inputs.

4. **A stock-like paycheck can take the whole stock budget.** If 40% of your future pay moves with the market, a balanced saver should hold 0% in stocks at 25, 24% at 35 and 33% at 55. The glide path runs upward, not downward.

5. **The 4% rule is obsolete in both directions.** A 30-year TIPS ladder bought today pays 4.91% of its cost a year, after inflation, with no chance of running out (my arithmetic from the Treasury real curve [1]). Taking 4% from a 60/40 portfolio runs out within 30 years in 19% of simulated paths at today's prices.

6 **Eleven asset classes go in and two come out.** After TIPS and global stocks, nothing else on the menu pays enough for the stock risk it carries. Nominal Treasuries lost their job when their correlation with stocks went from −0.38 to +0.59. Private assets get a weight of zero and a liquidity budget.

My take: at these prices the question is not which risky assets to own. It is how much of your wealth is still future savings, and whether that future pay is safe. Hold global stocks at market weights for the risk you want, hold TIPS that mature when you need the money for everything else, and move from the first to the second faster than a target-date fund does. My own portfolio fails that test on one count: it is 90% U.S. inside its stocks. The last section is the plan to fix it.

1. The question

What is the right portfolio? Most answers are a rule of thumb: 60/40, your age in bonds, a target-date fund, 4% a year in retirement. Each rule was fitted to a century of U.S. returns in which stocks earned about 8% a year more than cash. Nobody can buy that century. What you can buy is today's prices, and today's prices are published.

So this paper builds the answer from prices, in five steps. First, read expected real returns off the Treasury curve and off index valuations as of November 2026. Second, measure risk on the years since 2022, when real yields reset and stocks and bonds began falling together. Third, solve the portfolio problem for one investor with a stated risk aversion. Fourth, add the asset most models leave out, the savings a person has not made yet, which is what makes the answer depend on age and on the kind of job. Fifth, price retirement off the same curve. Then I simulate the whole path, sort the answers into a matrix of 45 groups, and hold my own portfolio up against it.

I take positions where the numbers support them. Where a result rests on an assumption I chose, I say so and show what happens if it is wrong.

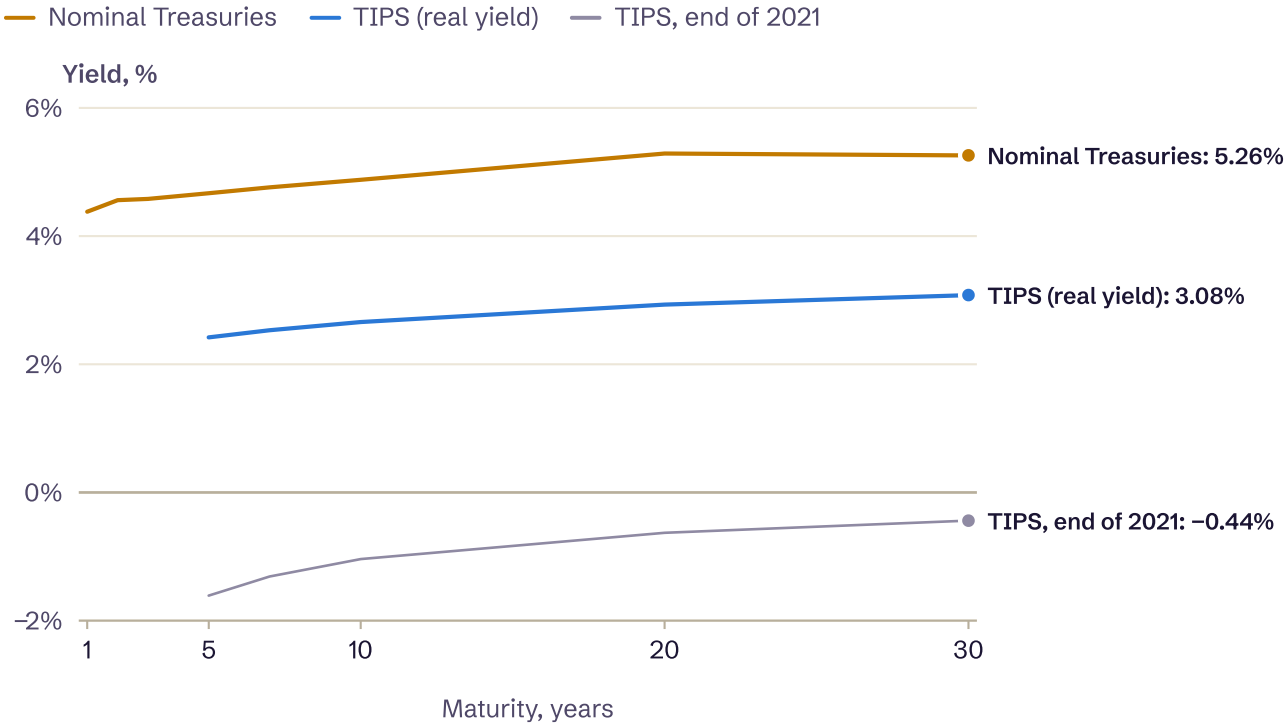
2. What prices say in November 2026

The safe asset

Treasury Inflation-Protected Securities pay a fixed yield on top of inflation. On November 9, 2026 the Treasury's par real curve stood at 2.42% for five years, 2.66% for ten and 3.08% for thirty [1]. At the end of 2021 the ten-year real yield was −1.04% [1]. Figure 1 shows both curves. An investor who needs purchasing power at a known date can now lock in close to 3% a year above inflation, with no credit risk and no forecast. Every other asset in this paper is measured against that.

Figure 1. The safe asset pays again

Treasury par yields by maturity on November 9, 2026, with the real curve at the end of 2021 for comparison



Breakeven inflation is the nominal yield minus the real yield at the same maturity (my arithmetic).

Source: U.S. Treasury, daily par real yield curve and par yield curve rates [1, 2].

Nominal Treasuries yield 4.88% at ten years [2]. The gap to TIPS, 2.22 points, is the inflation rate at which the two break even (my arithmetic). I use that breakeven as expected inflation, which makes the expected real return on a nominal Treasury equal to the TIPS yield at the same maturity. A nominal bond then offers the same expected return as TIPS plus inflation risk. That matters later.

Stocks

A stock index is a claim on earnings. Two standard ways to turn its price into an expected real return are the earnings yield, which assumes retained earnings earn their cost, and the dividend yield plus growth. I average them:

$$E[r_{stocks}] = \tfrac{1}{2} \cdot \tfrac{1}{2} \left(E/P_{trailing} + E/P_{forward}\right) + \tfrac{1}{2} \cdot (D/P + g)$$

E/P: earnings yield of the index on trailing and on forecast earnings. D/P: dividend yield. g: long-run real growth in earnings per share, 1.85% a year, the growth between the 1871-1880 average and the 2014-2024 average of real S&P earnings in Shiller's data [9] (my arithmetic).

These are compound returns: what a holder should expect to earn a year over a long horizon. A yield plus a compound growth rate cannot be anything else. I show compound returns in the tables, because that is what a saver ends up with. The portfolio formulas and the simulation need something different, the arithmetic mean of single-year returns, which is higher by about half the variance of those returns. For global stocks that is 1.14 points, so the

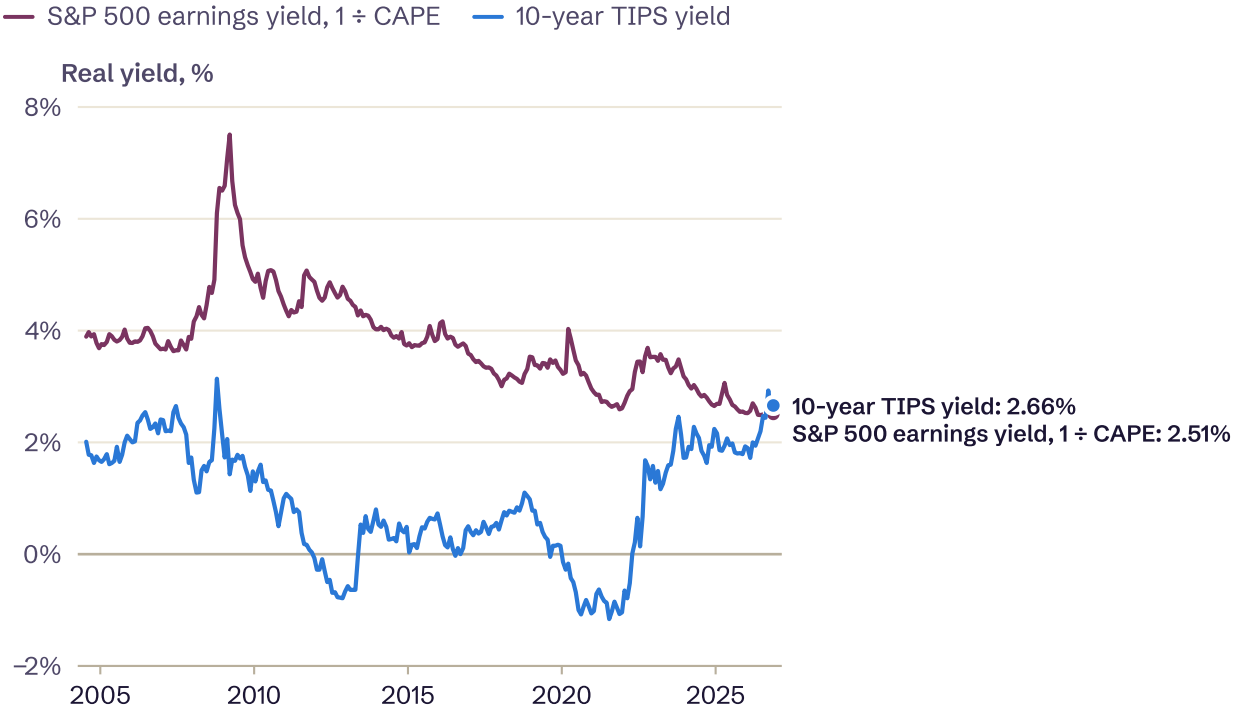
model uses 5.75% where the table shows 4.61%. Skipping that step would understate every stock share in this paper by about a third.

For the S&P Composite 1500 on November 6, 2026 the price was 22.4 times trailing earnings and 19 times forecast earnings, with a dividend yield of 1.18% [4]. The rule gives 3.95% a year after inflation. Developed markets outside the U.S. trade at 15.3 and 13.02 times with a 2.55% yield, which gives 5.76% [5]. Emerging markets give 5.68% [6]. At global market weights, 62.9% U.S., 27.5% other developed and 9.6% emerging [17], global stocks are priced to compound at 4.61%. All of these are my arithmetic from the cited index statistics.

No single method deserves much trust, so here are two others for the U.S. number. Shiller's cyclically adjusted P/E was 39.85 on November 9 [10], an earnings yield of 2.51%, which is 0.15 points below the ten-year TIPS yield. Damodaran's implied equity risk premium on November 1 was 3.79% over a 4.94% Treasury bond, an expected nominal return of 8.73% [11], or 6.51% after breakeven inflation (my arithmetic). My 3.95% sits between the two. One input pulls it up: the trailing P/E of 22.4 on the index page [4] is lower than the 25.75 that multpl reports on as-reported S&P 500 earnings [10]. Using the higher multiple would take 0.15 points off the U.S. return. Figure 2 shows why the low end deserves respect: for most of the last twenty years stocks out-yielded TIPS by three to six points. They no longer do.

Figure 2. Stocks used to out-yield TIPS by three to six points. Now they do not.

S&P 500 earnings yield on ten-year average real earnings (1 ÷ CAPE) and the 10-year TIPS yield, monthly



The gap is what Shiller calls the excess CAPE yield. CAPE is the first-of-month value; the TIPS yield is the last trading day of the month. The gap is my arithmetic.

Source: Shiller CAPE as published by multpl.com [10], following Shiller [9]; U.S. Treasury real yield curve [1].

Everything else

Credit pays the Treasury yield plus a spread, less what defaults take back. Investment-grade corporate bonds yield 5.74% with a spread of 90 basis points [7]. Over 150 years spreads have run about twice default losses [14], so I keep half the spread: 2.95% real. High-yield bonds yield 8.17% with a spread of 337 basis points [8]. The long-run default rate on high-yield bonds is 3.61% a year [15]; with 40% recovered, which is my assumption, the expected loss is 2.17% and the expected real return 3.53%.

Listed real estate yields 3.77% in dividends [13]. Since 1971 the price-only index of equity REITs has grown 3.85% a year against inflation of 3.92%, so real growth is −0.07% [13, 20] and the expected real return is 3.70% (my arithmetic). Gold has no yield, and its real price has tended to fall back after running above its average [16]; I give it zero. Cash is a four-week bill at 3.96% [3] less breakeven inflation: 1.71%.

Table 1. Real returns implied by prices on November 9, 2026

Each row is a yield or a valuation read off a public source, turned into a compound real return by the rule in the text

Asset class	What the price says	Priced real return, compound	Over 10-year TIPS	Realized, compound	Arithmetic mean the model uses
TIPS, 10-year	Real yield 2.66% [1]	2.66%	0.00	none before 1997	2.66%
Global stocks	U.S. 62.9%, international 27.5%, emerging 9.6% [17]	4.61%	+1.95	6.78% (U.S., 1928-2025)	5.75%
— U.S. stocks	P/E 22.4 trailing, 19 forward; dividend yield 1.18% [4]	3.95%	+1.29	7.16% (1926-2026)	-
— International developed	P/E 15.3, 13.02; yield 2.55% [5]	5.76%	+3.10	3.50% (1990-2026)	-
— Emerging markets	P/E 15.12, 13.53; yield 2.51% [6]	5.68%	+3.02	6.29% (1989-2026)	-
Listed real estate	Dividend yield 3.77%; real growth of the price index −0.07% a year since 1971 [13]	3.70%	+1.04	6.64% (1972-2026)	5.48%
High-yield credit	Yield 8.17%, spread 337 bp, duration 3.05 [8]; expected loss 2.17% [15]	3.53%	+0.87	no long history	3.85%
Investment-grade credit	Yield 5.74%, spread 90 bp, duration 6.45 [7]; expected loss 0.45% [14]	2.95%	+0.29	3.49% (Baa, 1928-2025)	3.31%
Long Treasuries	20-year yield 5.29% less breakeven 2.36% [1, 2]	2.93%	+0.27	1.45% (10-year, 1928-2025)	3.71%
Intermediate Treasuries	5-year yield 4.67% less breakeven 2.25% [1, 2]	2.42%	−0.24	1.45% (10-year, 1928-2025)	2.58%
Cash	4-week bill 3.96% less 5-year breakeven 2.25% [1–3]	1.71%	−0.95	0.33% (bills, 1928-2025)	1.72%
Gold	No yield; the real price reverts toward its mean [16]	0.00%	−2.66	2.50% (1928-2025)	1.71%

Index statistics are as of November 6, 2026; the Treasury curves as of November 9, 2026; the REIT yield is October 2026. Stocks: half the average of trailing and forward earnings yield, half dividend yield plus 1.85% real growth. Credit: Treasury real yield at the index's duration, plus spread, less expected loss. Realized returns are compound annual real returns from Damodaran [12], French [18] and Nareit [13]. The last column adds half of each asset's 2022-2026 variance to the priced return; TIPS held to maturity have none. The three stock regions are held as one asset. Every derived value is my arithmetic.

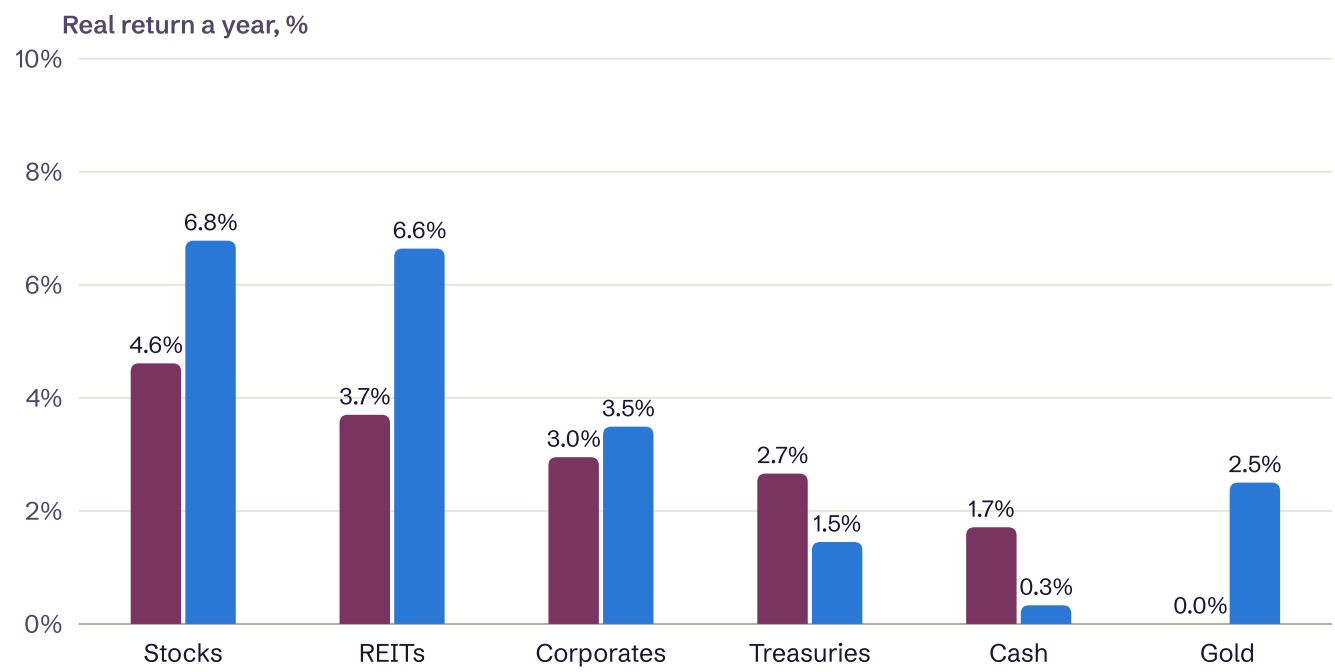
Source: As cited in each row.

Figure 3 puts these next to the averages that most allocation advice still runs on. From 1928 to 2025 the S&P 500 compounded at 6.78% a year after inflation (8.61% as an arithmetic average), ten-year Treasuries at 1.45% and bills

at 0.33% (my arithmetic from Damodaran's annual series [12]). Stocks are priced to compound at about 68% of that rate. Safe assets are priced to return more than they did. Compounded, the gap between stocks and the safe asset has gone from 6.45 points over bills to 1.95 over TIPS. As arithmetic means it has gone from 8.21 to 3.09.

Figure 3. What prices imply now, against the averages most advice is built on

Compound real return a year by asset class: implied by November 2026 prices, and realized



Implied: global stocks at market weights; listed real estate (FTSE Nareit All Equity REITs); investment-grade corporate bonds; Treasury bonds at the 10-year TIPS yield; a 4-week bill less breakeven inflation; gold at zero. Realized: compound annual real returns, 1928–2025: S&P 500, Baa corporates, 10-year Treasuries, 3-month bills, gold; real estate is my calculation from monthly FTSE Nareit returns, 1972–2026. Both sets are compound rates, so they compare like for like. All derived values are my arithmetic from the cited data.

Source: Table 1 for the implied values; Damodaran [12] and Nareit [13] for the realized returns.

3. Risk: the years since 2022

Expected returns come from prices. Risk has to come from data, and the choice of years is a choice of regime. I use monthly real returns from January 2022 to September 2026, 57 months. That window starts when real yields began to rise from below zero and covers the period in which stocks and Treasuries have moved together. It is short, so I shrink it and I show the longer windows beside it.

The series are public and none is built on an index from my employer. U.S. stocks are the CRSP market return and international stocks are French's developed ex-U.S. and emerging series [18]. Treasuries and TIPS are par bonds repriced each month from the Treasury curves [1, 2]. Real estate is the FTSE Nareit All Equity REITs index [13]. Credit and gold are fund net asset values [19]. Everything is deflated by the CPI [20]; the BLS published no index for October 2025, and I fill that month on a constant growth path.

$$\hat{\Sigma} = \delta \cdot F + (1 - \delta) \cdot S$$

S: sample covariance of monthly real returns, annualized. F: the same variances with every correlation set to the average correlation. δ: the shrinkage intensity that minimizes expected squared error (Ledoit and Wolf [21]). On the 2022–2026 window δ is 23% and the average correlation is 0.60.

Table 2. Risk inputs: volatility and correlation with U.S. stocks, by window

Annualized, from monthly real returns

Asset class	Volatility, 2022–26	2011–26	2011–21	Correlation with U.S. stocks, 2022–26	2011–21	Excess kurtosis
TIPS, 10-year	8.8%	6.9%	5.7%	+0.71	+0.07	2.2
U.S. stocks	16.3%	14.8%	14.1%	1.00	1.00	0.9
International developed stocks	15.8%	14.8%	14.4%	+0.80	+0.87	1.0
Emerging-market stocks	17.1%	16.9%	16.8%	+0.68	+0.75	1.0
Listed real estate	18.9%	16.8%	15.7%	+0.78	+0.68	0.9
High-yield credit	8.0%	7.5%	7.3%	+0.85	+0.77	4.5
Investment-grade credit	8.5%	6.4%	5.0%	+0.72	+0.32	2.3
Intermediate Treasuries	5.6%	4.2%	3.4%	+0.51	−0.32	0.6
Long Treasuries	12.5%	11.8%	11.4%	+0.59	−0.38	−0.1
Gold	18.5%	16.9%	16.1%	+0.15	+0.08	0.5
Cash	1.6%	1.3%	1.1%	+0.28	−0.01	0.4

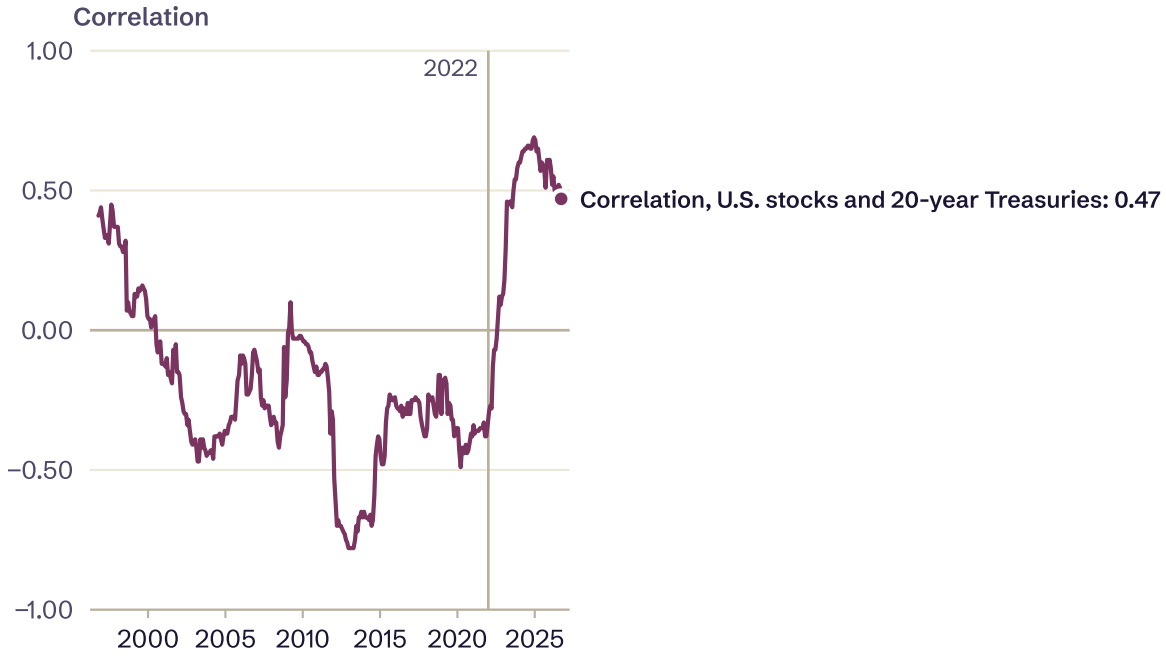
Windows: January 2022 to September 2026 (57 months), May 2011 to September 2026 (185), May 2011 to December 2021 (128). Unshrunk sample statistics; the model uses the 2022–26 matrix shrunk 23% toward a common correlation of 0.60 [21]. TIPS and Treasuries are par bonds repriced monthly from the Treasury curves; credit is fund NAV without distributions; gold is the NAV of a bullion trust. All my calculation.

Source: French [18]; U.S. Treasury [1, 2]; Nareit [13]; fund NAV histories [19]; BLS CPI [20].

Two things in Table 2 drive the results. First, volatility is higher than in the calm decade before: global stocks at market weights have run at 15.3% a year since 2022 against 13.8% from 2011 to 2021, and a ten-year TIPS bond at 8.8% against 5.7%. Second, the sign of the stock-bond correlation flipped. From May 2011 to December 2021 the correlation between U.S. stocks and a 20-year Treasury was −0.38. Since January 2022 it has been +0.59.

Figure 4. The hedge that left: stocks and long Treasuries now fall together

Rolling 36-month correlation of monthly returns, U.S. stocks and a 20-year Treasury bond



Nominal monthly returns. The bond is a 20-year par bond repriced each month from the Treasury curve (my construction). Source: Ken French data library (CRSP) [18]; U.S. Treasury par yield curve [2].

Figure 4 shows the rolling version back to the 1990s. The 36-month correlation averaged −0.31 from 2000 to 2020, turned positive in August 2022 and was 0.47 in September 2026. For twenty years a long Treasury was insurance that paid a yield. In this regime it is a second risky asset with the expected return of the safe one.

Monthly returns also have fatter tails than a normal distribution: excess kurtosis from 2011 to 2026 is about 0.9 for stocks, 2.2 for TIPS and 4.5 for high yield. The simulations below draw from a Student-t distribution with 5 degrees of freedom, not a normal one.

4. One investor, no paycheck

Start with someone who lives off a portfolio and nothing else. I give investors constant relative risk aversion, the standard assumption: losing a given percentage hurts the same at any level of wealth, and a single number γ says how much.

$$U(W) = W^{1-\gamma} / (1 - \gamma)$$

W: real wealth. γ: relative risk aversion. Higher γ means a bigger premium is needed to accept the same risk.

With one risky asset, one safe asset and continuous rebalancing, Merton and Samuelson showed in 1969 that such an investor holds a constant share in the risky asset, whatever the horizon [22, 23]:

$$\alpha^* = (\mu - r) / (\gamma\sigma^2)$$

α*: share of wealth in the risky asset. μ − r: its arithmetic mean return over the safe asset. σ²: its variance.

The safe asset is the one that is safe at the investor's horizon. For someone funding spending years away, that is an inflation-indexed bond maturing when the money is needed, not cash [25]. So r is the TIPS yield, and TIPS held to maturity carry no risk in the model even though their price moves in between. That is the one place I depart from a textbook mean-variance setup, and it matters: marked to market, a ten-year TIPS fund lost about 29% after inflation between late 2021 and late 2023 (my calculation from the Treasury real curve), while a TIPS bond held to maturity paid exactly what it promised. One simplification: I use the ten-year TIPS yield as r at every age. Against the thirty-year yield of 3.08% the balanced share below would be 29%, and against the five-year yield of 2.42% it would be 37%.

Risk aversion needs a scale people recognize. I use the old averages for that. With the 1928–2025 premium of 8.21 points over bills and volatility of 19.3%, a γ of 3.7 gives the classic 60% in stocks and a γ of 2.2 gives 100%. So I call γ = 2 aggressive, 3 growth, 4 balanced, 6 moderate, 8 cautious and 10 very cautious. The labels depend on what the old safe asset was. Against ten-year Treasuries instead of bills the old premium is 6.78 points, and 60% in stocks is a γ of 3.0, my growth investor. So read "the 60/40 investor" as γ between 3 and 4. Table 3 gives the share each holds on the old averages and on today's prices.

Table 3. The Merton share, today and under the averages

Stock share of a stand-alone investor with no future pay: π ÷ (γσ²)

Risk aversion γ	Group	Stocks, November 2026 inputs	Stocks, 1928–2025 averages
2	Aggressive	67.9%	110% (borrows)
3	Growth	45.2%	73.4%
4	Balanced	33.9%	55.1%
6	Moderate	22.6%	36.7%
8	Cautious	17.0%	27.5%
10	Very cautious	13.6%	22.0%

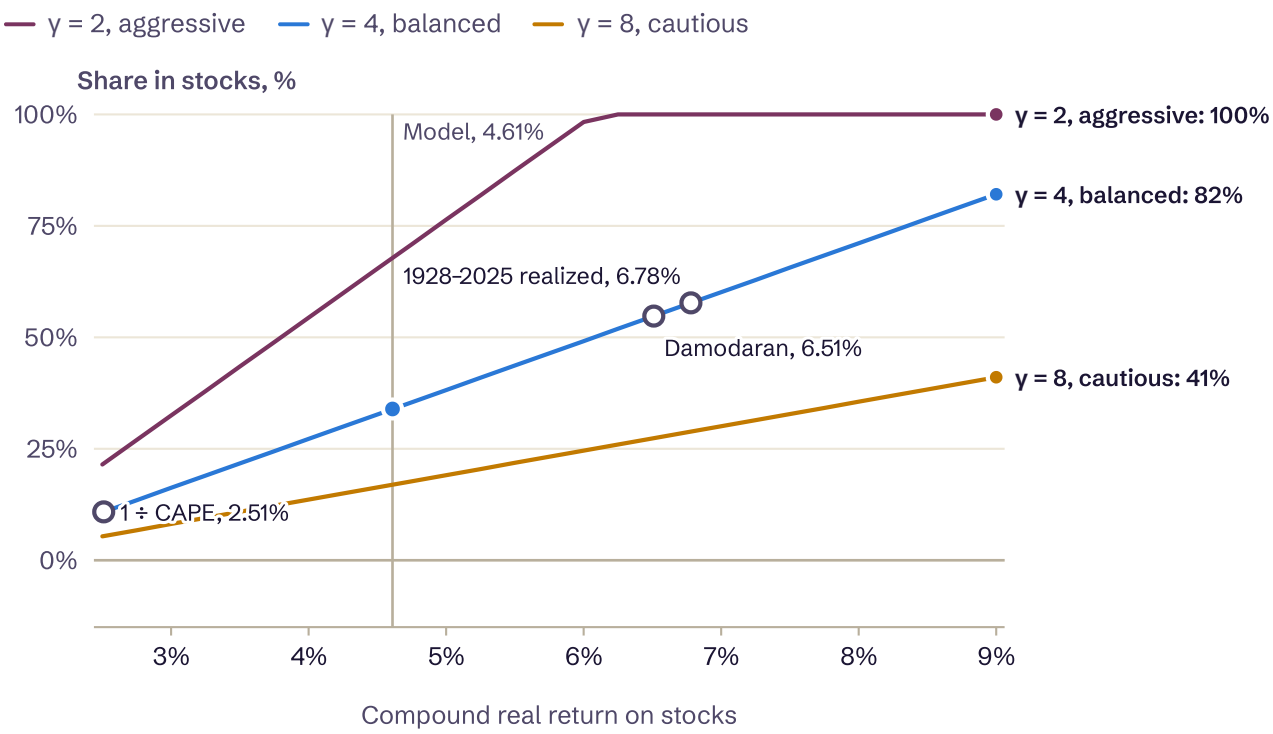
Today: premium 3.09 points over 10-year TIPS, volatility 15.1%. Averages: premium 8.21 points over bills, volatility 19.3% (arithmetic real returns, S&P 500 and 3-month bills). My arithmetic.

Source: Merton (1969) [22]; Table 1 and Table 2; Damodaran [12].

The balanced investor who held 55% in stocks on the averages holds 33.9% on today's prices. The growth investor goes from 73% to 45.2%, and the aggressive one from more than everything to 67.9%. Nothing about these people changed. The premium fell by 62% and the formula is linear in it; lower volatility gave some of that back.

Figure 5. The stock share is the premium divided by risk aversion times variance

Merton share of a stand-alone investor as the compound real return on stocks changes; TIPS at 2.66%, stock volatility 15.09%



Shares are capped at 100% (no borrowing). The share uses the arithmetic mean, the compound return on the axis plus half the variance. The points sit on the γ = 4 line; the vertical line marks the stock return this paper uses. My arithmetic.

Source: Merton (1969) [22]; inputs from Table 1 and Table 2.

Figure 5 shows how much rides on the stock return, the least certain input. Reading each estimate, like mine, as a compound return: at the CAPE yield the balanced investor holds 11% in stocks, and at Damodaran's number 55%. Only the top of that range gets her near 60%. It also shows how little is at stake. Holding the Merton share instead of no stocks at all is worth 52 basis points a year in certainty-equivalent return to the balanced investor today, against 2.3 points on the old averages ($\pi^2 \div 2\gamma\sigma^2$, my arithmetic). When the safe asset pays 2.66%, getting the stock share exactly right matters about a quarter as much as it did on the old averages.

One more number puts the premium in terms a saver feels. Compounding 1.95 points faster than TIPS with volatility of 15.1%, global stocks have about a 34% chance of ending behind TIPS over ten years and 21% over forty. For U.S. stocks alone, at 1.29 points and 16.3% volatility, it is 40% and 31% (lognormal approximation, my arithmetic).

5. Eleven asset classes go in, two come out

The menu has eleven asset classes: TIPS; U.S., international developed and emerging-market stocks; listed real estate; investment-grade and high-yield credit; intermediate and long Treasuries; gold; and cash. A private-asset sleeve is handled separately below. I hold the three stock regions at global market weights as one asset, for a reason I come back to. That leaves nine. The investor's problem, with no borrowing and no shorting, is

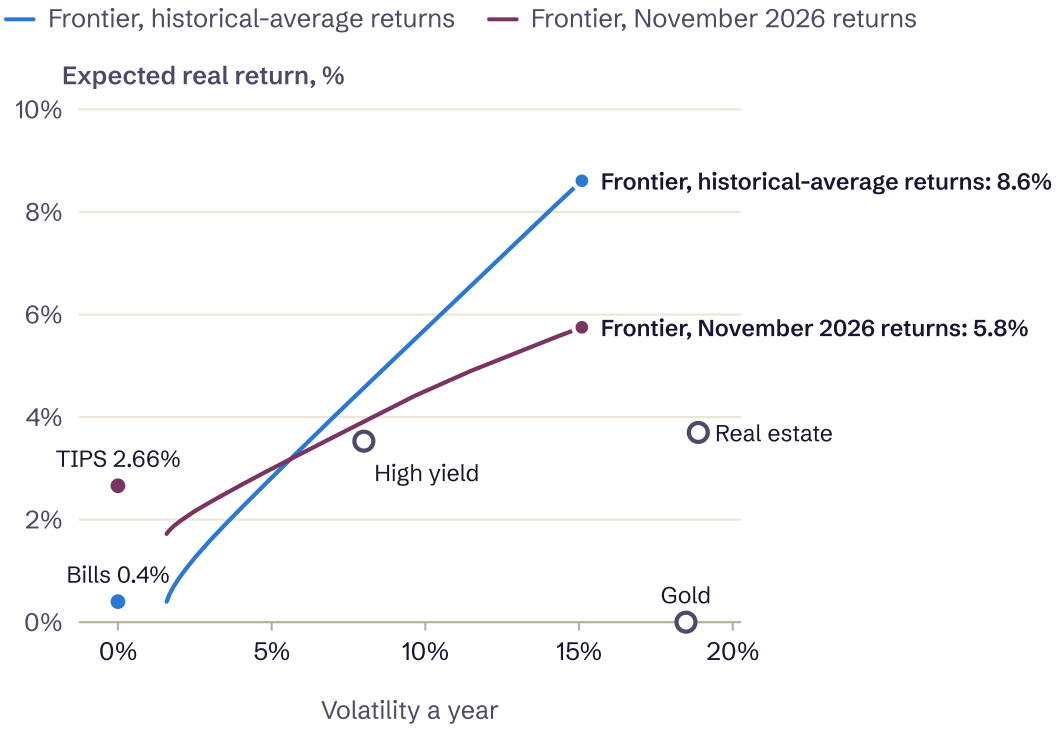
$$\max_x \mu'x - (\gamma/2)\, x'\Sigma x \text{ subject to } x \geq 0,\ 1'x = 1$$

x: portfolio weights. μ: arithmetic mean real returns, the last column of Table 1. Σ: the shrunk 2022-2026 covariance, with the TIPS row and column set to zero.

This is a quadratic program. With nine assets it can be solved exactly by brute force: for each of the 511 possible sets of assets held, the first-order conditions and the budget constraint are a small linear system; the answer is the solution with no negative weight and the highest objective. I checked the weights with a second implementation in another language on a covariance matrix rebuilt from the raw downloads.

Figure 6. Same risks, 38% of the reward

Efficient frontier of the eight risky asset classes, long-only, with November 2026 implied returns and with historical-average returns; risk is the 2022-2026 covariance in both



Each curve is the best expected return available at each level of risk from the eight risky asset classes. The filled points at zero volatility are the safe asset in each case; the slope from there to the best risky mix is 0.544 with historical averages and 0.205 today. Hollow points are single asset classes at today's implied returns. High yield has no long history, so its historical return uses Baa corporates. My calculation.

Source: Table 1 and Table 2; historical averages from Damodaran [12] and Nareit [13].

Figure 6 shows what the optimizer is choosing among. With today's returns the best mix of risky assets has a slope of 0.20 over TIPS. With historical-average returns and the same risks it is 0.54 over bills. The line an investor can travel along is 38% as steep as it was.

Table 4. What each asset class adds, once stocks and TIPS are on the menu

Stand-alone investor, long-only optimum over all nine asset classes

Asset class	Beta to global stocks	Premium left after beta, bp	Weight at γ = 2	Gain in certainty-equivalent return, bp a year
Global stocks	1.00	-	67.9%	40.9
Listed real estate	0.99	–24	0.0%	0
High-yield credit	0.44	–17	0.0%	0
Investment-grade credit	0.43	–68	0.0%	0
Intermediate Treasuries	0.23	–79	0.0%	0
Long Treasuries	0.55	–65	0.0%	0
Gold	0.45	–233	0.0%	0
Cash	0.04	–107	0.0%	0

"Premium left after beta" is the asset's premium over TIPS minus its beta times the stock premium. The gain is the certainty-equivalent return with the asset available minus without it, at the γ on the grid where it is largest. The model keeps an asset only if it adds at least 1 basis point a year. My calculation.

Source: Model output; inputs from Table 1 and Table 2.

Table 4 asks what each asset adds once TIPS and stocks are available. The test is its premium over TIPS after subtracting its beta times the stock premium. Listed real estate has a beta of 0.99 and a premium 24 basis points too small for it. High yield is 17 short, investment-grade credit 68, long Treasuries 65, gold 233. Nothing clears zero, so the optimizer holds none of them at any risk aversion on the grid. The model's rule is to keep an asset only if it adds at least one basis point a year of certainty-equivalent return, and none adds anything. High yield and real estate are the closest calls: 17 and 24 basis points are inside the error of my estimates, and I would not argue with someone who held a little of either. Two assets are left: TIPS and global stocks.

That is a statement about November 2026 prices, not about these asset classes. Table 5 reruns the problem with risk measured on other years. On the 2011–2021 covariance, when long Treasuries hedged stocks, the same compound returns give long Treasuries 36.6% of the portfolio and let the balanced investor hold 49.1% in stocks. With only stocks and TIPS she would hold 39.1% on that window's lower volatility, so the hedge was worth about 10 points of stock exposure. It is gone, and the model does not pay for insurance that stopped paying.

Table 5. Same priced returns, different years of risk data

Long-only optimum for a stand-alone balanced investor (γ = 4) under each covariance estimate

Risk estimated on	Stock-long-bond correlation	Stock volatility	Stocks	Long Treasuries	High yield	TIPS	Stocks if only stocks and TIPS
2022-2026, shrunk (the model)	+0.66	15.1%	33.9%	0.0%	0.0%	66.1%	33.9%
2022-2026, no shrinkage	+0.66	15.3%	33.5%	0.0%	0.0%	66.5%	33.5%
2011-2026, shrunk	+0.02	14.1%	36.9%	16.3%	0.0%	46.8%	37.2%
2011-2021, shrunk	−0.32	13.5%	49.1%	36.6%	0.0%	14.3%	39.1%

All nine asset classes are available in these runs. Compound returns are the same in every row; each row adds half of its own variances to get arithmetic means. The correlation is global stocks against 20-year Treasuries. The last column is the share with only stocks and TIPS on the menu, so the difference from the Stocks column is what the other assets add. My calculation.

Source: Model output.

Why the regions are not optimized

If I let the optimizer choose among the three stock regions at their own implied returns, it puts nothing in the U.S.: a balanced investor gets 28.2% international developed, 18.4% emerging and no U.S. stocks at all. That is what a 1.8-point gap in expected return does when two assets have a correlation of 0.80. I do not believe my regional estimates that much. The gap rests on earnings multiples, a valuation gap can stay open for years, and a corner solution turns any error in it into the whole answer. So the model lets prices set the regional mix, at market weights, and lets valuations set only how much stock to own. The practical reading is narrower and I am confident in it: a portfolio that is all U.S. stocks is betting against the half of the valuation evidence that is clearest.

6. The asset nobody puts on the statement

A 30-year-old with one year of pay saved does not have a small portfolio. She has a small portfolio plus thirty-five years of contributions that have not happened yet. If her job is secure, those contributions are close to a bond: they arrive on schedule whatever the stock market does. Bodie, Merton and Samuelson made the point in 1992 [24], and Cocco, Gomes and Maenhout put it in one line: labor income substitutes for riskless asset holdings, so the optimal stock share falls with age [26].

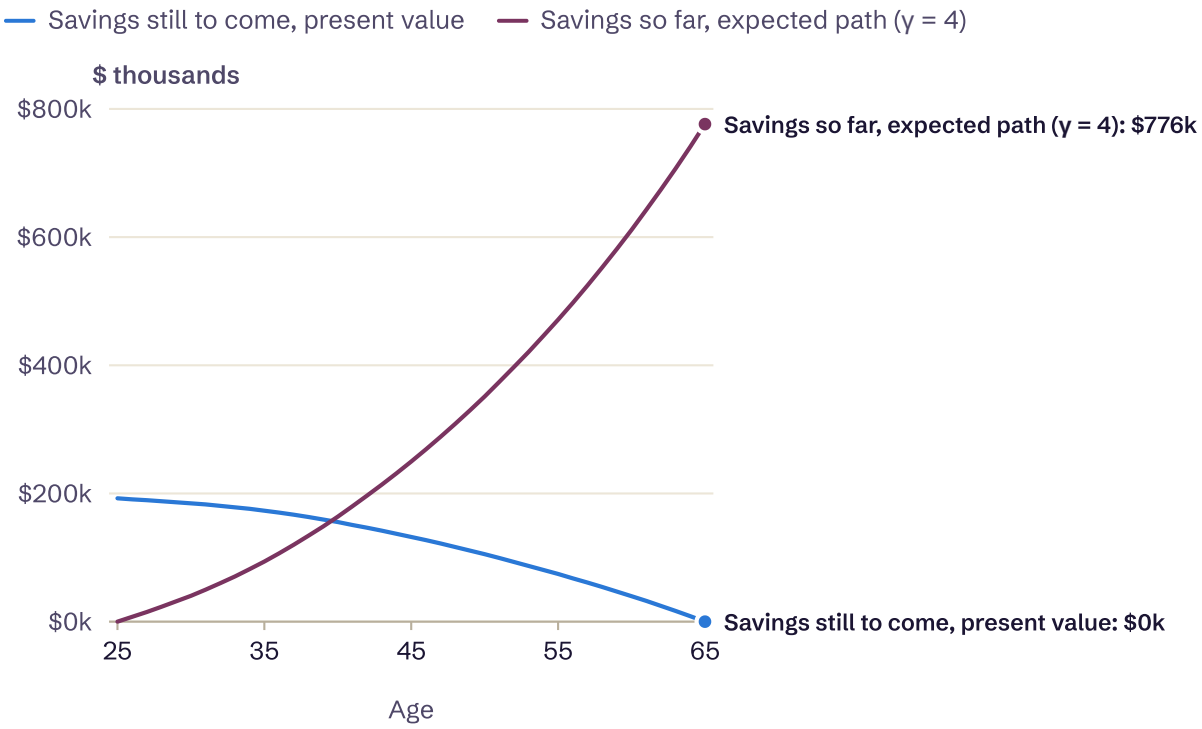
I count only the part of future pay that will be saved. The rest pays for life before retirement and is not available to absorb a market loss unless the saver is willing to cut spending. With a fixed saving rate, the present value of future contributions is exactly the wealth that has yet to arrive in the portfolio:

$$H_a = \Sigma_{t=a}^{R-1} s \cdot Y_t \cdot S_{a,t} / (1 + z_{t-a} + \beta \cdot \pi)^{t-a+½}$$

H_a: human capital at age a. s: share of pay saved, 12.1% including the employer's contribution [30]. Y_t: median pay at age t for full-time, year-round workers [28], held flat in real terms. $S_{a,t}$: probability of surviving from a to t [31]. z: the real Treasury yield at that horizon [1]. β: the share of future pay that behaves like stocks. π: the stock premium. R: retirement age, 65.

Figure 7. For the first fifteen years, most of the retirement pot has not been earned yet

Savings so far and the present value of savings still to come, for a median full-time worker saving 12.1% of pay from 25; thousands of 2025 dollars



Pay is the 2025 median for full-time, year-round workers by age, held flat in real terms. Savings to come are weighted by survival and discounted on the real Treasury curve, plus β times the equity premium when pay is stock-like. Savings so far grow at the model portfolio's expected return. My calculation.

Source: Census Bureau CPS ASEC [28]; SSA life table [31]; Treasury real curve [1]; Vanguard savings rate [30].

Total retirement wealth is savings so far, F, plus H. Write k = F / (F + H). The investor wants the Merton share of total wealth in stocks, and a share β of H already behaves like stocks. The portfolio problem on total wealth is the same quadratic program with two changes:

$$\max_x \mu' x - (\gamma/2)\,[\,k \cdot x'\Sigma x + 2(1-k) \cdot \beta \cdot x'\Sigma e_{stocks}\,] \text{ subject to } x \geq 0,\ 1'x = 1$$

x: weights on savings. e_{stocks}: the vector that picks out global stocks. The first term in brackets scales risk down because savings are only a share k of wealth. The second charges the portfolio for risk the paycheck already carries.

With two assets left on the menu the solution has a closed form, and it is the whole model in one line:

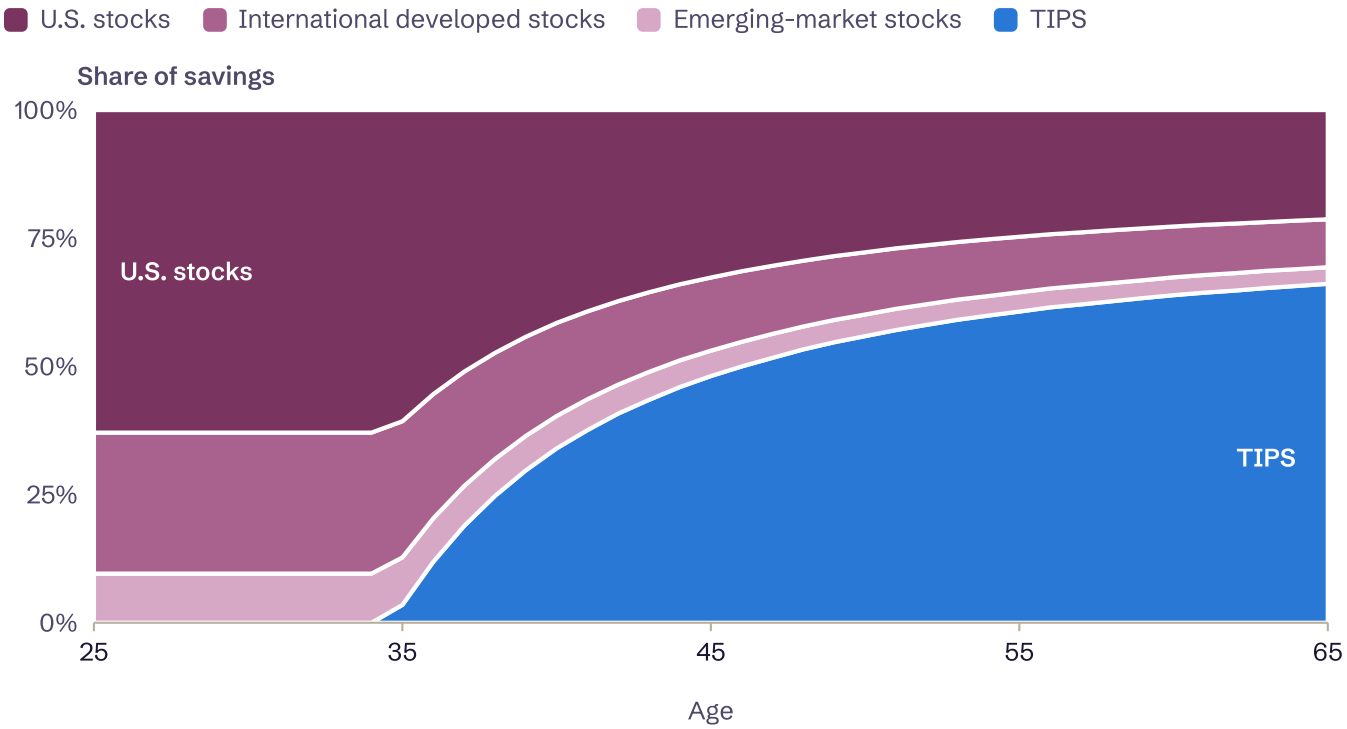
$$\textit{stock share of savings} = \textit{clip}\left(\left[\pi / (\gamma\sigma^2) - \beta(1 - k) \right] / k \, , \, 0, \, 1 \right)$$

π: arithmetic stock premium over TIPS, 3.09 points. σ: stock volatility, 15.1%. clip keeps the share between 0 and 100%: no borrowing, no shorting.

Read it from the inside out. $\pi / (\gamma\sigma^2)$ is the Merton share, the stock budget as a fraction of total wealth. $\beta(1 - k)$ is how much of that budget the paycheck has already spent. Dividing by k turns a share of total wealth into a share of savings, which is why a young saver with little in the bank can be 100% in stocks while holding only a few percent of total wealth in them.

Figure 8. The glide path the math produces: all stocks to 34, then a fast walk to TIPS

Model portfolio by age for a balanced investor (γ = 4) with stable pay, percent of savings



Stocks are held at global market weights. This is the expected path; after a fall in stocks, savings shrink relative to the savings still to come and the stock share goes up, not down. My calculation.

Source: Model output; FTSE Global All Cap weights [17].

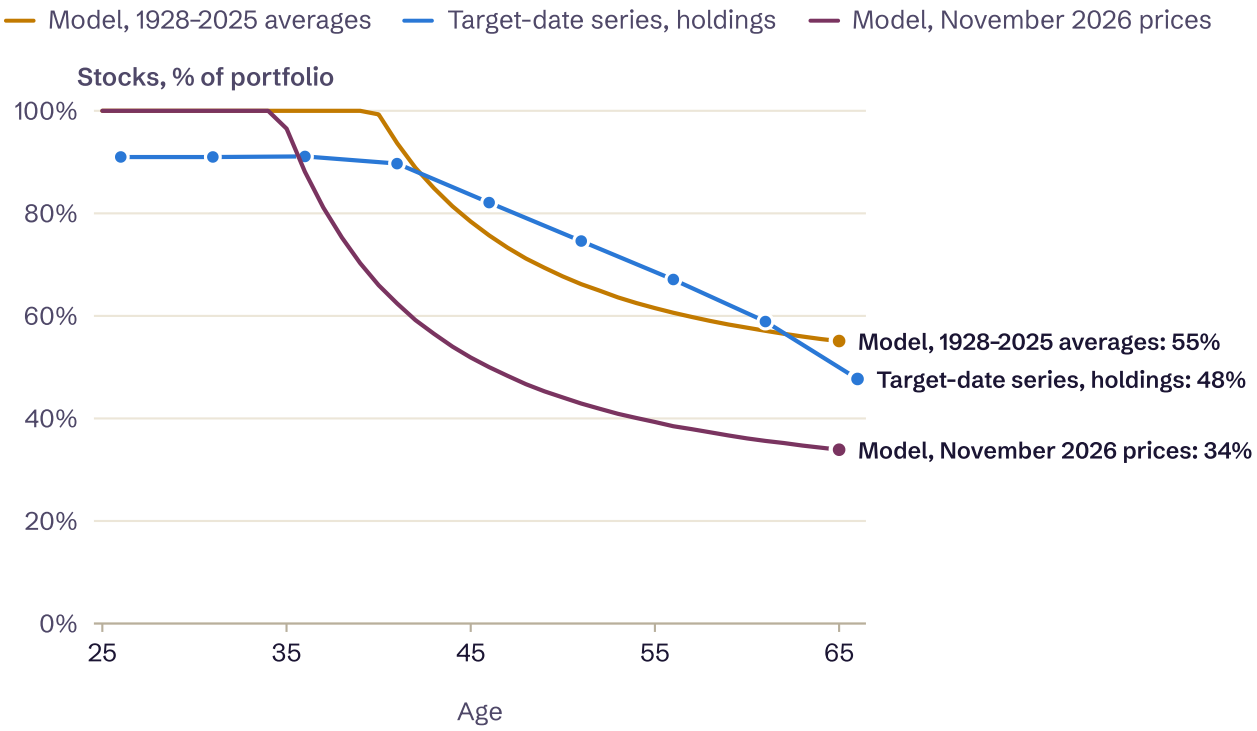
Figure 8 is the result for a balanced saver with stable pay: all stocks until 34, 96.5% at 35, 66.0% at 40, 44.1% at 50 and 33.9% at 65. I did not choose that shape. It is the formula evaluated along the path of a median earner who saves 12.1% of pay. An aggressive saver stays all in stocks until 45 and ends at 67.9%. A cautious one is at 94.4% by 30 and ends at 17.0%.

This is where target-date funds come from

Target-date funds hold 93% in stocks at the median for savers 45 years from retirement [33]. The largest series, with 38% of target-date assets [35], holds 91.1% for a 36-year-old, 74.6% at 51, 58.9% at 61 and 47.7% at 66, going by its funds' reported holdings [32]. Feed my model the 1928–2025 averages instead of today's prices and it produces 99.3% at 40, 67.7% at 50 and 55.1% at 65. That is a target-date fund. Figure 9 draws the three together.

Figure 9. A target-date fund is this model run on the old averages

Stock share by age for a balanced investor (γ = 4, stable pay): the model at November 2026 prices, the same model fed 1928–2025 averages, and the largest target-date series



Target-date points are issuer-reported fund holdings as of September 30, 2026, placed at the age of an investor retiring at 65 in the vintage year; the 2025 fund (age 66) holds 47.7% and the income fund 31.2%. "Averages" uses the 1928–2025 premium of 8.21 points over bills and 19.3% volatility. The model lines stop at 65; in retirement the stock share depends on the funded ratio (Table 7).

Source: Vanguard Target Retirement fund holdings, as republished by StockAnalysis [32]; Damodaran [12]; model output.

So the industry's glide path is not wrong in its logic. It is right for a premium of 8.21 points. At 3.09 points the same logic says a balanced saver should be at 66.0% in stocks at 40, not 90%, and at 33.9% at 65, not about half. The fund is one path sold to everyone, and it does not fit everyone the same way. A growth saver (γ = 3) ends at 45.2%, about where the fund does. An aggressive one should hold more stock than the fund from her forties on: 87.1% at 50 and 67.9% at 65. What none of them should copy is the other half of the fund, nominal bonds held in a fund, where the model holds TIPS that mature when the money is needed.

When the paycheck is the risky asset

Not everyone's pay is a bond. Benzoni, Collin-Dufresne and Goldstein show that when wages and dividends move together over long horizons, a young worker's human capital is stock-like and the right stock share is hump-shaped over life, not falling [27]. I use three groups: stable pay with β of 0, cyclical pay with β of 0.15 (bonus-heavy or cyclical industries), and stock-like pay with β of 0.4 (commission, founders, pay that is largely equity). These β values are my assumptions, chosen to span that literature, not estimates.

Figure 10. Stock share by age, risk aversion and how stock-like the paycheck is

Model stock share, percent of savings, along each group's expected path; the rest is TIPS

0 100 Rows: risk aversion

Stable pay (β = 0)

	25	30	35	40	45	50	55	60	65
γ 2 Aggressive	100	100	100	100	100	87	78	72	68
γ 3 Growth	100	100	100	88	69	58	52	48	45
γ 4 Balanced	100	100	97	66	52	44	39	36	34
γ 6 Moderate	100	100	65	45	35	30	26	24	23
γ 8 Cautious	100	94	49	34	27	22	20	18	17
γ 10 Very cautious	100	76	39	27	21	18	16	15	14

Cyclical pay (β = 0.15)

	25	30	35	40	45	50	55	60	65
γ 2 Aggressive	100	100	100	100	94	82	75	71	68
γ 3 Growth	100	100	98	73	61	54	50	47	45
γ 4 Balanced	100	100	67	51	44	40	37	35	34
γ 6 Moderate	100	55	36	30	27	25	24	23	23
γ 8 Cautious	100	26	21	19	18	18	17	17	17
γ 10 Very cautious	0	7	11	12	13	13	13	14	14

Stock-like pay (β = 0.4)

	25	30	35	40	45	50	55	60	65
γ 2 Aggressive	100	100	100	91	81	75	72	70	68
γ 3 Growth	100	66	54	50	48	47	46	46	45
γ 4 Balanced	0	10	24	28	31	32	33	34	34
γ 6 Moderate	0	0	0	6	13	17	20	21	23
γ 8 Cautious	0	0	0	0	4	9	13	15	17
γ 10 Very cautious	0	0	0	0	0	5	9	12	14

Age

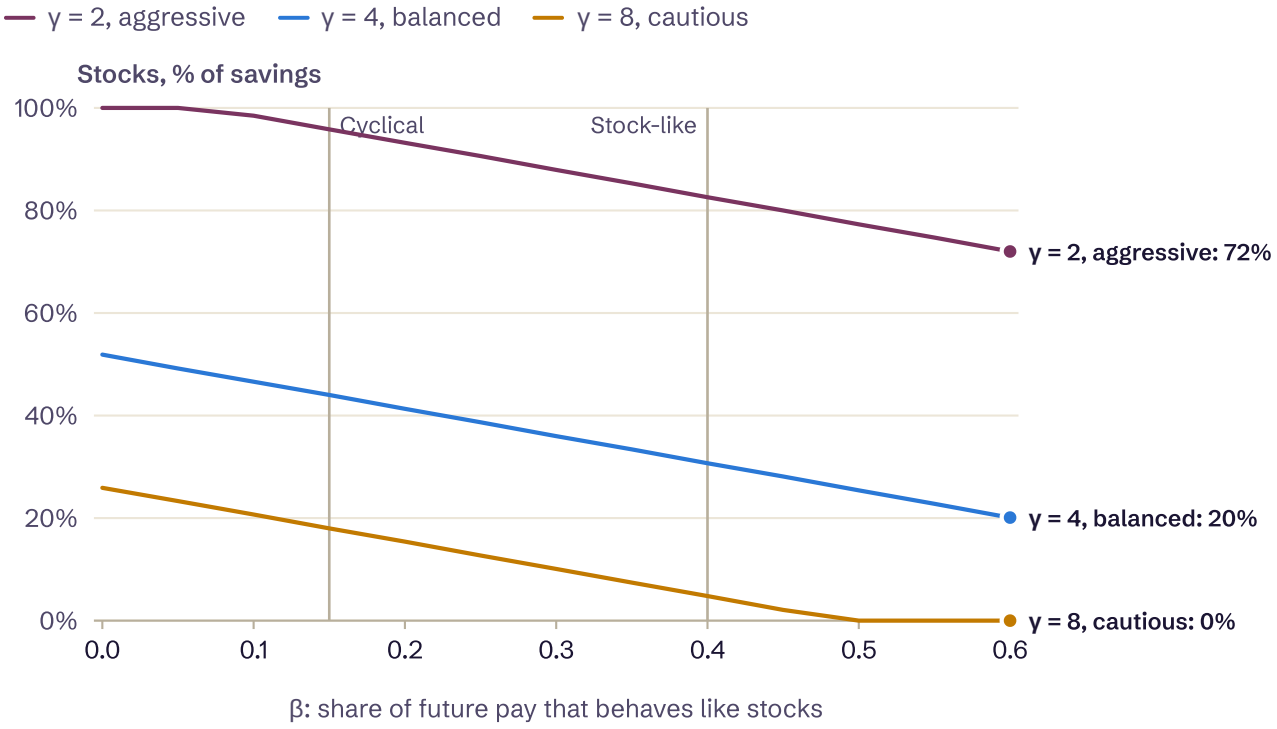
β is the share of future pay that behaves like stocks. Each row follows its own expected savings path, so cautious rows hold slightly less wealth at a given age. My calculation.

Source: Model output.

Figure 10 is the full grid. With cyclical pay a balanced saver holds 67% in stocks at 35, against 97% with stable pay. With stock-like pay she holds 0% at 25, 24% at 35 and 34% at 60: the glide path runs up. Her paycheck is the stock position, and it shrinks as she ages, so the portfolio takes over. An aggressive saver is different. Her stock budget of 68% is larger than the 40% her pay uses up, so even with stock-like pay she holds 100% at 30.

Figure 11. A stock-like paycheck uses up the stock budget

Model stock share at age 45 as the share of future pay that behaves like stocks (β) rises, by risk aversion



Savings so far are 65% of retirement wealth at 45 on the stable-pay path. At 30, where they are 18%, the lines are 9 times steeper once the no-borrowing limit stops binding: a balanced investor goes from 97.5% in stocks at β of 0.2 to 6.2% at 0.4. My calculation.

Source: Model output.

Figure 11 shows how sharp this is. At 45, each tenth added to β takes about 5 points off a balanced saver's stock share. At 30 the no-borrowing limit hides the effect until β passes 0.2; from there to 0.4 the share falls from 98% to 6%. For a young saver the most important input in this paper is not the stock return. It is an honest answer to the question of what happens to her income in a bad market.

One alternative deserves a line. If a saver would truly cut spending or work longer after a crash, all future pay counts, not only the saved part. Then H is about eight times larger and the balanced saver with stable pay stays all in stocks until 51 and then falls to the same 33.9% at 65. I use the savings-only version because it is the one the saver controls, and because the simulation below holds the saving rate fixed.

7. Retirement: buy the floor, then decide

At retirement the paychecks stop and the question changes from how much to accumulate to how much can be spent. The real curve answers it by arithmetic. A TIPS ladder is a set of bonds that mature one per year, each sized to pay for that year. Its price comes from discount factors bootstrapped off the par real curve:

$$d_n = (1 - c_n \cdot \Sigma_{j<n}\, d_j) / (1 + c_n),\ w_N = 1 / (1 + \Sigma_{n=1}^{N-1}\, d_n)$$

c_n: par real yield at n years, as an annual rate. d_n: price today of one real dollar in n years. w_N: the level real income, as a share of the purchase price, from N annual payments starting now.

Table 6. What the real curve funds, by length of retirement

Level real income from a TIPS ladder bought on November 9, 2026, first payment at purchase, for a buyer aged 65

Payments	Last payment at age	Income, % of cost a year	Chance a man is alive after the last payment	A woman	Either of a couple
20	84	6.40%	44.9%	56.6%	76.1%
25	89	5.49%	24.1%	34.9%	50.6%
30	94	4.91%	8.0%	14.4%	21.3%
35	99	4.50%	1.3%	3.2%	4.4%

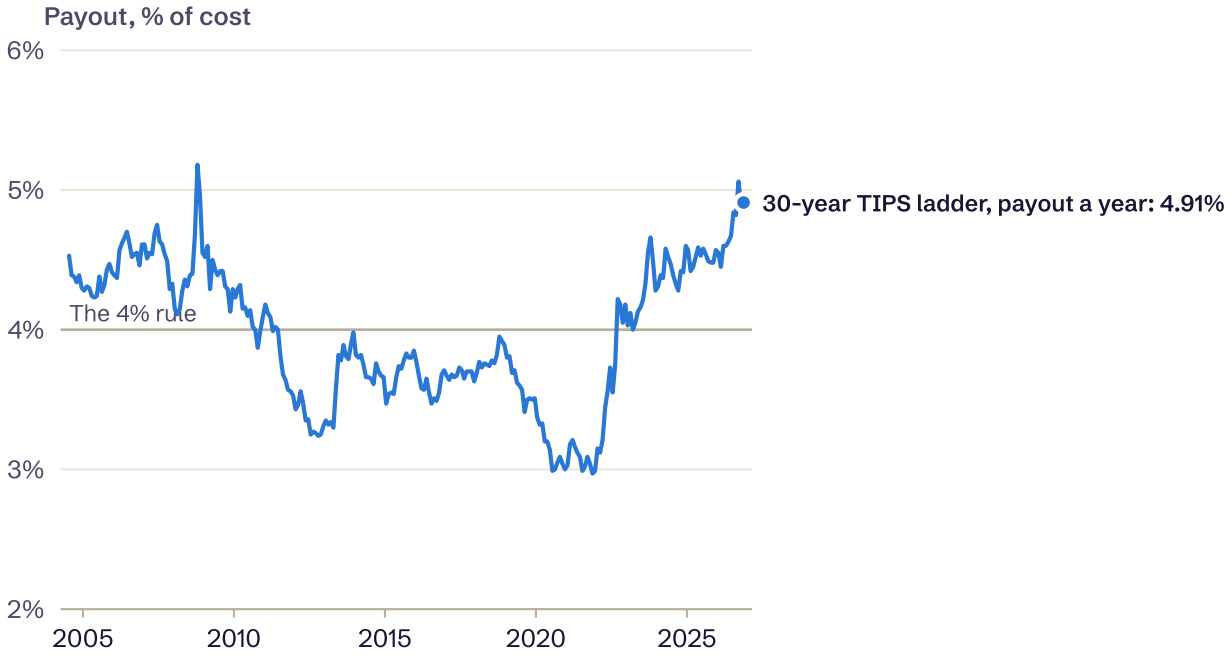
Survival is from age 65 to the birthday after the last payment, SSA 2023 period table; the couple column assumes independent lives. A fair real annuity for one 65-year-old would pay 6.77%, the average of the rates for a man and a woman (my arithmetic from the same table and curve, before any insurer's costs). My calculation.

Source: U.S. Treasury real yield curve [1]; SSA life table [31].

On the November 9 curve a ladder of thirty payments pays 4.91% of its cost a year, rising with inflation, from 65 through 94 (my arithmetic [1]). There is no simulation in that number and no probability of failure. Figure 12 shows what the same ladder paid at each month-end curve since 2004. In November 2021 it paid 2.97%. The 4% rule made sense when the safe alternative paid three.

Figure 12. A TIPS ladder now pays more than the 4% rule, with nothing left to chance

Real income for 30 years from a TIPS ladder bought at each month-end real curve, percent of purchase price a year



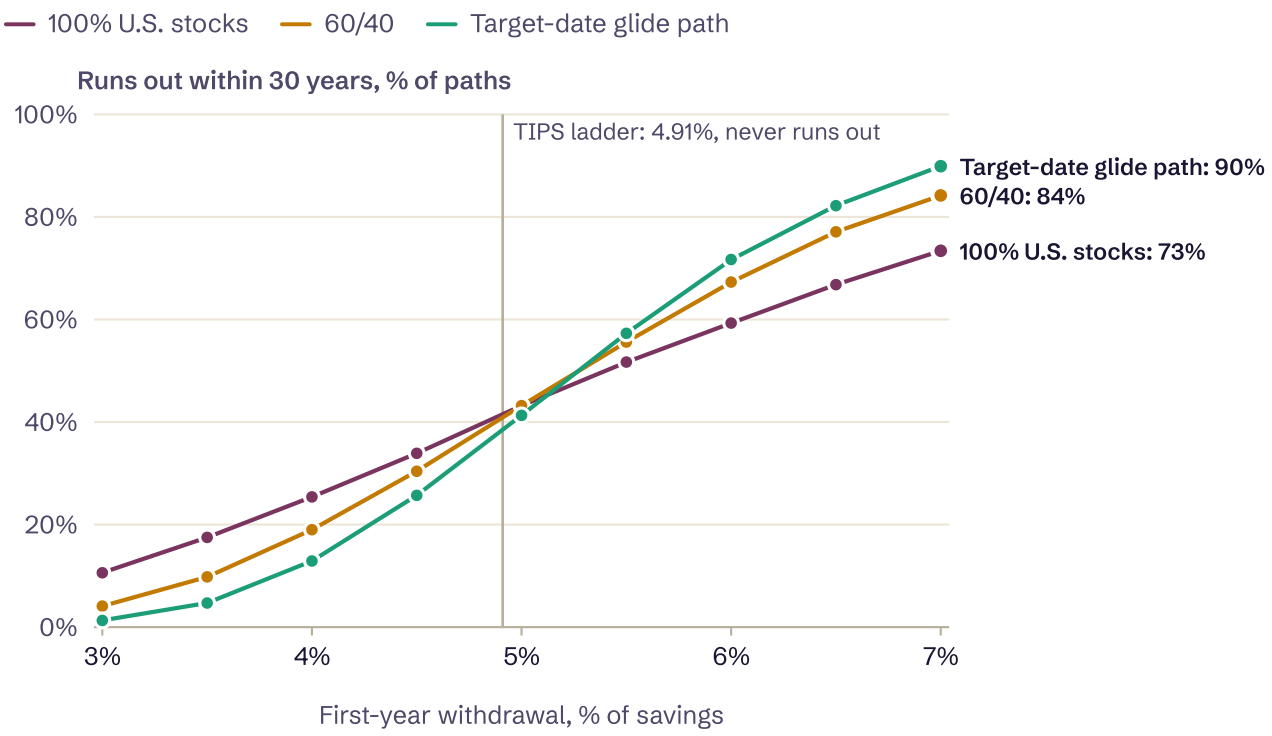
Thirty equal real payments, the first on the purchase date. Discount factors are bootstrapped from the par real curve (5 to 30 years; flat inside five years; the 20-year point extended before 2010). My calculation.

Source: U.S. Treasury par real yield curve [1].

The 4% rule comes from Bengen, who published it in 1994 as a worst case from U.S. history: take 4% of the portfolio in the first year, raise it with inflation, and the money lasted 30 years even for someone who retired at the 1968 peak [34]. At today's prices it is not a worst case. Figure 13 runs the test forward with November 2026 expected returns: at a 4% withdrawal a 60/40 portfolio runs out within 30 years in 19.0% of paths, the target-date glide path in 12.9% and an all-stock portfolio in 25.4%. The rule asks a risky portfolio to do, with a failure rate of 13% to 25%, what a ladder now does with certainty at a higher rate.

Figure 13. At today's prices the 4% rule fails in 13% to 25% of retirements

Share of 10,000 simulated 30-year retirements in which a fixed real withdrawal runs out, by first-year withdrawal rate



The withdrawal is set at 65 as a share of savings and then held constant in real terms. Monthly returns from a multivariate Student-t with five degrees of freedom, the November 2026 expected returns and the 2022–2026 covariance. 60/40 is U.S. stocks and intermediate Treasuries. The sampling error on each point is under half a point. My simulation.

Source: Model output; inputs from Table 1 and Table 2.

The floor and the surplus

A ladder has one weakness. It ends. A 65-year-old man has an 8% chance of being alive at 95, a woman 14%, and at least one of a couple 21% [31]. So the model splits retirement wealth in two. The floor is essential spending not covered by Social Security, funded by a ladder long enough for the household. The surplus is everything else. The funded ratio is wealth divided by the cost of the floor. If utility is defined over spending above the floor, the Merton result applies to the surplus alone:

$$\textit{stock share of wealth} = [\,\pi / (\gamma\sigma^2)\,] \cdot (1 - 1/FR) \textit{ for } FR \geq 1, \textit{ and } 0 \textit{ below}$$

FR: funded ratio. 1 / FR of wealth buys the floor. The Merton share of what is left goes to stocks; the rest of the surplus is also TIPS.

Table 7. Retirees by funded ratio

Funded ratio is savings divided by the cost of a TIPS ladder that pays for essential spending not covered by Social Security

Funded ratio	Ladder (the floor)	Stocks, y = 2	y = 4	y = 8	Expected real return, y = 4	Bad year 1 in 100, y = 2
0.80	100% (funds 80% of the floor)	0.0%	0.0%	0.0%	2.66%	2.7%
1.00	100.0%	0.0%	0.0%	0.0%	2.66%	2.7%
1.25	80.0%	13.6%	6.8%	3.4%	2.86%	–2.3%
1.50	66.7%	22.6%	11.3%	5.7%	3.00%	–5.5%
2.00	50.0%	33.9%	17.0%	8.5%	3.15%	–9.6%
3.00	33.3%	45.2%	22.6%	11.3%	3.30%	–13.7%

The floor goes into the ladder first; the surplus is invested at the stand-alone Merton share and the rest of the surplus is also TIPS. Below a funded ratio of 1 the model holds no stocks: the choice is a smaller floor, a later retirement or more saving, not more risk. My calculation.

Source: Model output.

Table 7 is the retiree half of the matrix. A balanced retiree with one and a half times the floor holds 11.3% in stocks. An aggressive retiree with three times the floor holds 45.2%, the only one in the table near the 48% that a target-date fund holds at 66 [32]. Below a funded ratio of one the model holds no stocks at all, and that is the result I expect people to like least. The usual instinct is that someone short of their goal needs growth. The math says that someone who cannot fund the floor cannot afford to make it smaller, and stocks that compound 1.95 points faster than TIPS are unlikely to close the gap: the choices are a smaller floor, more saving or a later date.

Many households are nearer that position than the model saver's. The median family headed by someone 55 to 64 had a net worth of $411,000 in the 2025 Survey of Consumer Finances, 4.5 times that group's median income [29]. The model saver who put away 12.1% from 25 has 8.6 years of pay at 60.

8. Forty years, ten thousand times

The formulas above are instantaneous. To see what they do over a life I simulate it: 10,000 paths, monthly, from 25 to 94. Each month's real returns for all eleven asset classes are drawn together:

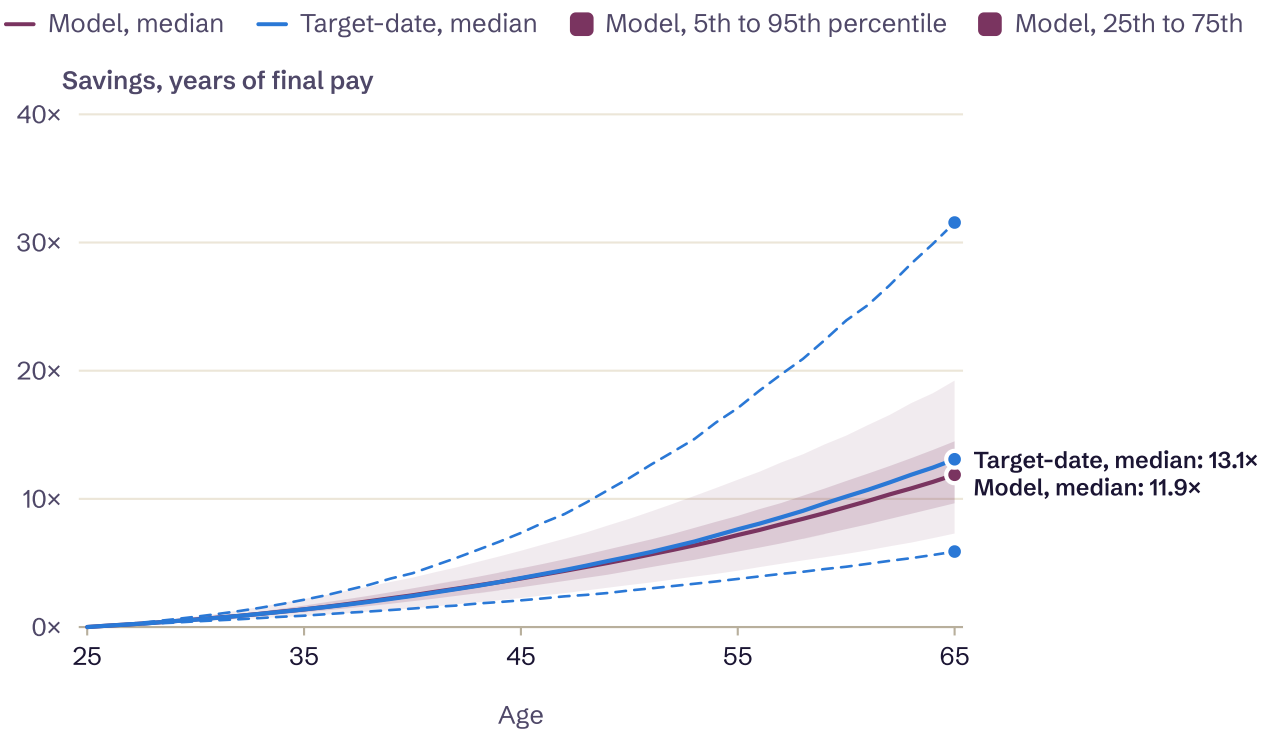
$$r_t = \mu/12 + L\, z_t \cdot \sqrt{((\nu - 2)/\nu)} \,/\, \sqrt{(\chi^2_{\nu,t} / \nu)}$$

μ: arithmetic mean real returns, the compound returns in Table 1 plus half of each asset's variance. L: Cholesky factor of the shrunk 2022-2026 covariance, divided by 12. z: independent standard normals. χ^2_ν: a chi-square draw shared by all assets in the month, which makes the result a multivariate Student-t with ν = 5 degrees of freedom and the right covariance. Bad months are worse and more common than under a normal distribution, and they hit every asset at once.

The saver earns the median pay for her age and saves 12.1% of it each month. With stock-like pay, her income moves each year by β times the surprise in stock returns, permanently. Five strategies face the same draws. The model re-solves its formula each year on actual savings and the present value of contributions still to come, holds TIPS that mature when the money is needed, buys a ladder with its TIPS at 65 and amortizes its stocks over the years left. Four benchmarks invest the same contributions: 60/40 in U.S. stocks and intermediate Treasuries; the target-date glide path of Figure 9; 100% U.S. stocks; and age in bonds. Each benchmark spends in retirement by amortizing what it holds over the years left.

Figure 14. Forty years of saving: the range matters more than the middle

Savings in years of final pay for a balanced saver (γ = 4, stable pay, 12.1% of pay): model portfolio, 10,000 paths



Dashed lines are the target-date path's 5th and 95th percentiles. Same simulation as Figure 13. Final pay is the median at 64. The model's TIPS are bought to mature when the money is needed, so they earn their real yield without price risk; the target-date fund's bonds are marked to market each month. My simulation.

Source: Model output.

Figure 14 shows the saving years for a balanced saver. The model's median is a little lower than the target-date fund's at 65, 11.9 years of final pay against 13.1, and its range is much narrower: 7.3 to 19.2 between the 5th and 95th percentiles, against 5.9 to 31.6. That is what the model is for. It gives up 9% of median wealth and its 5th-percentile outcome is 24% better.

Table 8. Forty years of saving and thirty of spending, five strategies

10,000 simulated lives, balanced investor (γ = 4) with stable pay saving 12.1% of median pay from 25

Strategy	Savings at 65, years of final pay: 5th / median / 95th	Income, % of final pay: 5th / median	Worst year, 5th pct.	Below TIPS-only	Certainty-equivalent income
Model	7.3 / 11.9 / 19.2	38% / 68%	32%	8%	53%
60/40	5.2 / 10.8 / 23.6	24% / 61%	15%	25%	34%
Target-date	5.9 / 13.1 / 31.6	29% / 72%	21%	16%	43%
100% stocks	4.2 / 12.1 / 39.7	19% / 74%	10%	24%	21%
Age in bonds	5.5 / 10.3 / 20.6	25% / 53%	19%	29%	38%
Model, TIPS in a 10-year fund	5.6 / 11.6 / 25.2	30% / 67%	25%	16%	45%
TIPS only	–	41% / 41%	41%	–	41%

Income is the average real retirement income from 65 to 94 as a share of pay at 64; Social Security is on top and not modeled. “Worst year” is the 5th percentile across lives of the lowest single year of income. “Below TIPS-only” is the share of lives in which average income ends below the TIPS-only saver’s. “TIPS only” saves in TIPS at today’s 10-year real yield and buys a ladder at 65: a certain outcome. 60/40 and 100% stocks hold U.S. stocks; target-date follows Figure 9; age-in-bonds holds its age in intermediate Treasuries. My simulation.

Source: Model output; inputs from Table 1 and Table 2.

Table 8 carries the same lives through retirement. A saver who bought nothing but TIPS for forty years and a ladder at 65 would replace 41% of final pay for thirty years before Social Security, if every contribution could be invested at today’s real yields. That is the bar. The target-date path beats it at the median, 72%, and falls short of it in 16% of lives. 60/40 falls short in 25% and all U.S. stocks in 24%. The model falls short in 8%.

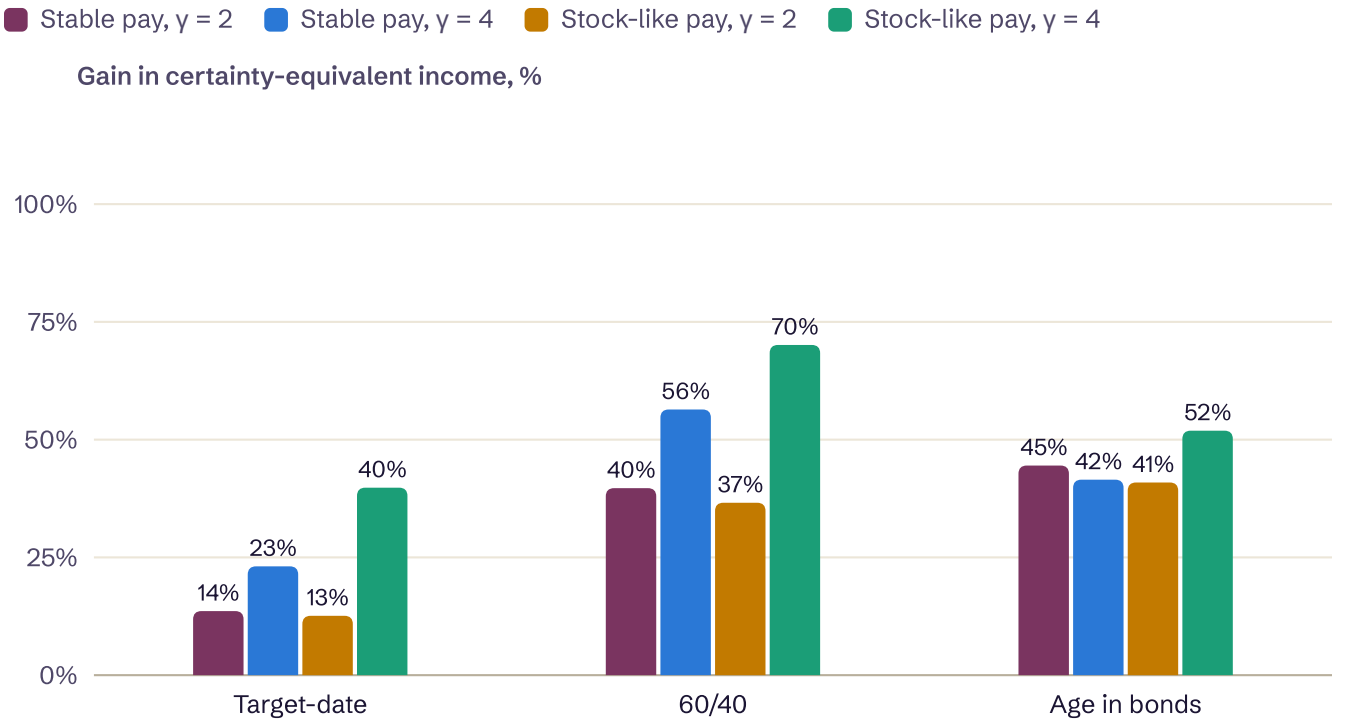
One assumption carries much of this, and I want it in plain sight. The simulation holds the real yield curve fixed. Every future contribution buys TIPS at today’s yields, and the ladder at 65 is priced on today’s curve. Only the TIPS a saver already owns are certain. Real yields were below zero five years ago, and TIPS are not issued beyond thirty years, so a saver under 35 cannot yet buy the bonds that would pay for her last years. If real yields fall, the paths that lean on TIPS end with less than the table shows.

$$CE = [\,\Sigma_t\, S_t \cdot E(c_t^{\,1-\gamma}) \,/\, \Sigma_t\, S_t\,]^{1/(1-\gamma)}$$

CE: certainty-equivalent income, the constant real income the investor would accept in place of the risky stream. c_t: real spending at age t, 65 to 94. S_t: probability of being alive [31]. The expectation is over simulated paths.

Figure 15. What the model is worth against the usual answers

Gain in certainty-equivalent retirement income from following the model instead of each benchmark, percent



Certainty-equivalent income is the constant real income a CRRA investor with that γ would accept in place of the simulated stream from 65 to 94, weighted by survival. Each strategy spends by amortizing what it holds over the years left; the model holds its TIPS as a ladder. My simulation.

Source: Model output; SSA life table [31].

Scored by certainty-equivalent income, the model is worth 23% more than the target-date path to a balanced saver and 14% more to an aggressive one (Figure 15). Three things produce that, and I would rather name them than let the number stand alone. The first is matching: the model's TIPS mature when the money is needed, so they carry no price risk, while the benchmarks hold bond funds. If the model held its TIPS in a ten-year fund instead, its edge over the target-date path for the balanced saver would shrink to 3%. So most of the edge is the matching, and the matching rests on the fixed-curve assumption above. The second is the stock mix, which matters against the two U.S.-only benchmarks and not against the target-date fund, whose stocks are close to global weights. The third is less stock after 40, which on its own is worth little. The gains over 60/40 are larger because 60/40 gets none of the three right.

9. Private assets: a budget, not a weight

Private equity, private credit and private real estate are the twelfth asset class, and the model gives them a default weight of zero. The reason is in the companion paper, The Illiquidity Premium Changed Hands [38]: after fees and an honest adjustment for leverage and smoothed prices, the average buyer of private funds has not been paid a premium for the lockup, and what premium exists goes to whoever supplies liquidity when others need it. In this model's terms, an asset with the expected return of public stocks, more volatility once its marks are unsmoothed, and a ten-year lock fails the test in Table 4 before the lock is even counted.

What the model does provide is the limit. An investor can supply liquidity only with money that will not be needed, even in a crash. The liquidity budget is the largest share of wealth that can be locked for ten years without forcing a sale of the rest at the bottom:

$$L = max(\,0,\ 1 - N / [\,W \cdot (1 - D \cdot s)\,]\,)$$

N: cash the investor must be able to raise over ten years (a reserve for savers; the first ten rungs of the ladder for retirees). W: wealth. s: share of the liquid portfolio in stocks. D: the fall in stocks to plan for, 53.6%, the worst real peak-to-trough loss in global stocks since 1990 (my calculation from [18]).

Table 9. The liquidity budget: the most that could be locked up for ten years

Share of savings an investor could lock without becoming a forced seller in a crash; an upper limit, not an allocation

Investor	Savings, years of pay	Cash needed over ten years, % of savings	Stocks in the liquid part	Liquidity budget	Model weight in private assets
Saver, age 30, stable pay	0.6	68%	100%	0%	0%
Saver, age 45, stable pay	3.3	13%	52%	82%	0%
Saver, age 60, stable pay	8.6	5%	36%	94%	0%
Saver, age 30, cyclical pay	0.6	136%	100%	0%	0%
Saver, age 45, cyclical pay	3.2	27%	44%	65%	0%
Saver, age 60, cyclical pay	8.3	11%	35%	87%	0%
Saver, age 30, stock-like pay	0.6	282%	10%	0%	0%
Saver, age 45, stock-like pay	3.0	58%	31%	30%	0%
Saver, age 60, stock-like pay	7.8	23%	34%	72%	0%
Retiree, funded ratio 1.00	-	44%	0%	56%	0%
Retiree, funded ratio 1.50	-	29%	11%	69%	0%
Retiree, funded ratio 2.00	-	22%	17%	76%	0%

Budget = 1 − need ÷ (1 − D × stock share), floored at zero, with D = 53.6%, the worst real fall in global stocks since 1990 (trough in February 2009; my calculation from [18]). Need: for savers, a reserve of 6, 12 or 24 months of spending for stable, cyclical and stock-like pay (my assumption); for retirees, the first ten rungs of the ladder, 44% of its cost. Savers are balanced investors (γ = 4) on their expected path.

Source: Model output.

Table 9 shows who has a budget. A 30-year-old does not: the reserve is most of what she owns. A 45-year-old saver with stable pay could lock up to 82% and a 60-year-old 94%, because they are adding money, not drawing it. A retiree who has just funded the floor could lock 56% by this arithmetic, but all of it is committed to the ladder, so the usable budget is the surplus. The people with room to supply liquidity are mid-career savers with secure

pay. The budget is a ceiling. The weight stays at zero until a specific fund clears the hurdle in the companion paper's calculator.

10. The matrix

Here is the model sorted into groups: five ages, three levels of risk aversion and three kinds of paycheck, each on its own expected path. Figure 16 shows the balanced investor; Table 10 has all 45 with expected real return, volatility, a bad year and the reason each differs from its neighbors. Retirees are in Table 7, by funded ratio.

Figure 16. The model portfolios for a balanced investor, by age and paycheck

Percent of savings for γ = 4; compound expected real return and volatility a year at right

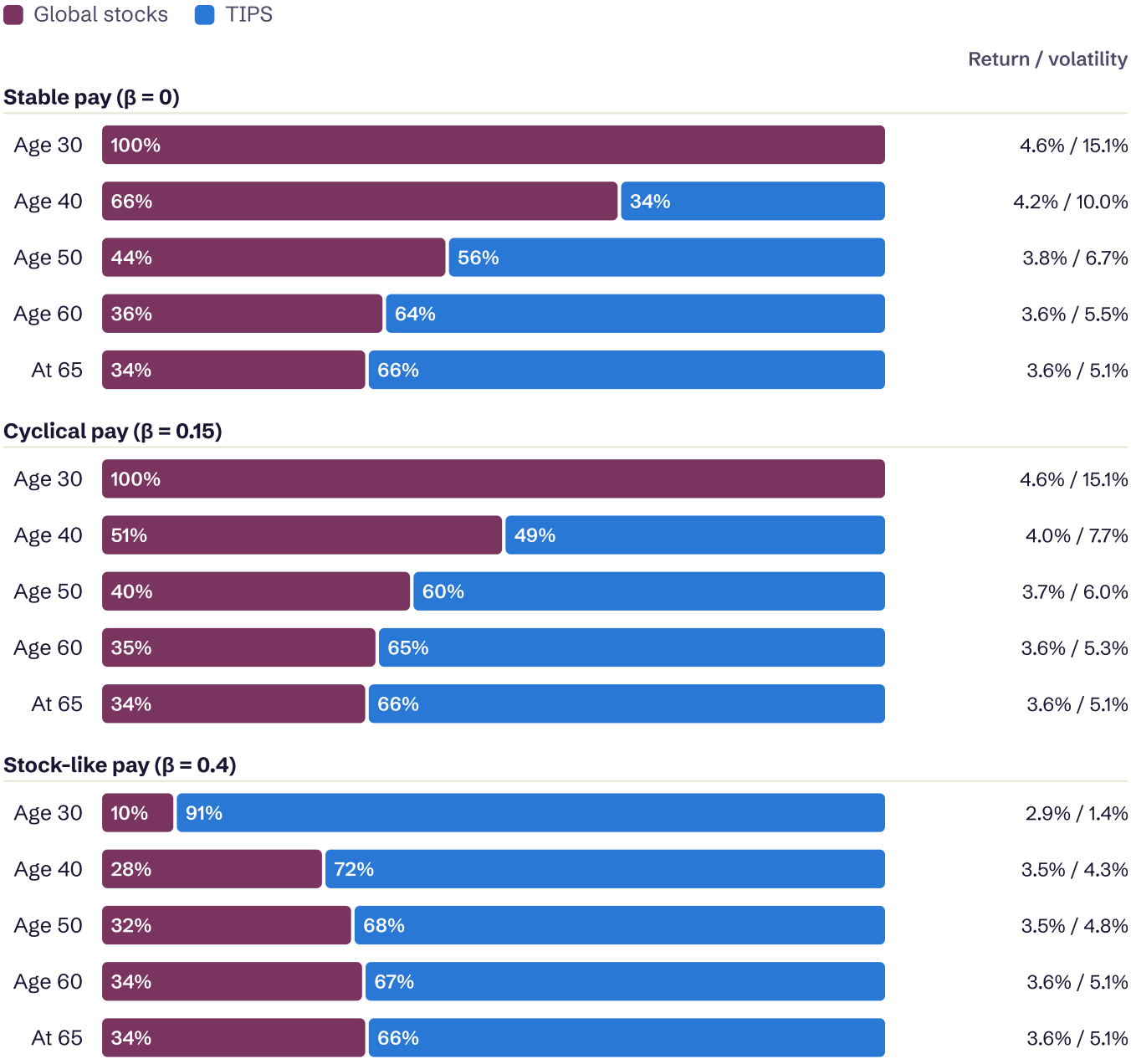


Table 10 has all 45 groups. Volatility counts TIPS as held to maturity.

Source: Model output.

Table 10. The matrix: model portfolios for 45 groups

By age, risk aversion and kind of paycheck: percent of savings in global stocks and in TIPS, with expected real return and three measures of risk. The line under each row says why it differs from its neighbors

Risk	Pay	Savings, years of pay	Stocks	TIPS	Expected real return, compound	Volatility	Bad year, 1 in 100	Worst fall 2011-26
AGE 25 TO 34, SHOWN AT 30								
Aggressive, y 2	Stable	0.7	100%	0%	4.61%	15.1%	–33.6%	–30.1%
Future savings are worth 4.6 times the portfolio and act like a bond. The no-leverage limit binds.								
Aggressive, y 2	Cyclical	0.7	100%	0%	4.61%	15.1%	–33.6%	–30.1%
Future savings are worth 4.3 times the portfolio. The no-leverage limit binds.								
Aggressive, y 2	Stock-like	0.7	100%	0%	4.61%	15.1%	–33.6%	–30.1%
Future savings are worth 3.9 times the portfolio. The no-leverage limit binds.								
Balanced, y 4	Stable	0.7	100%	0%	4.61%	15.1%	–33.6%	–30.1%
Future savings are worth 4.6 times the portfolio and act like a bond. The no-leverage limit binds.								
Balanced, y 4	Cyclical	0.7	100%	0%	4.61%	15.1%	–33.6%	–30.1%
Future savings are worth 4.3 times the portfolio. The no-leverage limit binds.								
Balanced, y 4	Stock-like	0.6	10%	90%	2.94%	1.4%	–0.8%	–1.7%
The paycheck uses up 32 points of the 34% stock budget; the portfolio holds the rest.								
Cautious, y 8	Stable	0.7	94%	6%	4.56%	14.2%	–31.5%	–28.5%
Future savings are 82% of retirement wealth, so the portfolio holds 5.6 times the stand-alone 17% stock share.								
Cautious, y 8	Cyclical	0.6	26%	74%	3.38%	3.9%	–6.6%	–7.0%
The paycheck uses up 12 points of the 17% stock budget; the portfolio holds the rest.								
Cautious, y 8	Stock-like	0.6	0%	100%	2.66%	0.0%	2.7%	0.0%
The paycheck already supplies 32% of retirement wealth in stock risk, more than the 17% this investor wants.								
AGE 35 TO 44, SHOWN AT 40								
Aggressive, y 2	Stable	2.2	100%	0%	4.61%	15.1%	–33.6%	–30.1%
Future savings are worth 0.9 times the portfolio and act like a bond. The no-leverage limit binds.								
Aggressive, y 2	Cyclical	2.2	100%	0%	4.61%	15.1%	–33.6%	–30.1%
Future savings are worth 0.9 times the portfolio. The no-leverage limit binds.								
Aggressive, y 2	Stock-like	2.2	91%	9%	4.53%	13.7%	–30.3%	–27.5%
The paycheck uses up 18 points of the 68% stock budget; the portfolio holds the rest.								
Balanced, y 4	Stable	2.2	66%	34%	4.20%	10.0%	–21.3%	–20.1%

Risk	Pay	Savings, years of pay	Stocks	TIPS	Expected real return, compound	Volatility	Bad year, 1 in 100	Worst fall 2011-26
Future savings are 49% of retirement wealth, so the portfolio holds 1.9 times the stand-alone 34% stock share.								
Balanced, y 4	Cyclical	2.1	51%	49%	3.95%	7.7%	–15.9%	–15.4%
The paycheck uses up 7 points of the 34% stock budget; the portfolio holds the rest.								
Balanced, y 4	Stock-like	2.0	28%	72%	3.45%	4.3%	–7.6%	–7.9%
The paycheck uses up 19 points of the 34% stock budget; the portfolio holds the rest.								
Cautious, y 8	Stable	2.1	34%	66%	3.57%	5.1%	–9.5%	–9.7%
Future savings are 50% of retirement wealth, so the portfolio holds 2 times the stand-alone 17% stock share.								
Cautious, y 8	Cyclical	2.0	19%	81%	3.20%	2.9%	–4.2%	–4.7%
The paycheck uses up 7 points of the 17% stock budget; the portfolio holds the rest.								
Cautious, y 8	Stock-like	1.9	0%	100%	2.66%	0.0%	2.7%	0.0%
The paycheck already supplies 20% of retirement wealth in stock risk, more than the 17% this investor wants.								
AGE 45 TO 54, SHOWN AT 50								
Aggressive, y 2	Stable	4.9	87%	13%	4.49%	13.1%	–28.9%	–26.4%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Aggressive, y 2	Cyclical	4.9	82%	18%	4.43%	12.4%	–27.2%	–25.0%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Aggressive, y 2	Stock-like	4.9	75%	25%	4.34%	11.4%	–24.6%	–22.8%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Balanced, y 4	Stable	4.7	44%	56%	3.80%	6.7%	–13.3%	–13.1%
Savings dominate; the mix is close to the stand-alone optimum of 34% stocks.								
Balanced, y 4	Cyclical	4.6	40%	60%	3.70%	6.0%	–11.7%	–11.7%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Balanced, y 4	Stock-like	4.2	32%	68%	3.53%	4.8%	–9.0%	–9.2%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Cautious, y 8	Stable	4.4	22%	78%	3.30%	3.4%	–5.5%	–5.9%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								
Cautious, y 8	Cyclical	4.1	18%	82%	3.17%	2.7%	–3.7%	–4.3%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								
Cautious, y 8	Stock-like	3.9	9%	91%	2.94%	1.4%	–0.7%	–1.6%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								
AGE 55 TO 64, SHOWN AT 60								
Aggressive, y 2	Stable	9.5	72%	28%	4.29%	10.8%	–23.4%	–21.8%

Risk	Pay	Savings, years of pay	Stocks	TIPS	Expected real return, compound	Volatility	Bad year, 1 in 100	Worst fall 2011-26
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Aggressive, y 2	Cyclical	9.4	71%	29%	4.28%	10.7%	–23.1%	–21.5%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Aggressive, y 2	Stock-like	9.3	70%	30%	4.26%	10.5%	–22.5%	–21.1%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Balanced, y 4	Stable	8.6	36%	64%	3.63%	5.5%	–10.4%	–10.5%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Balanced, y 4	Cyclical	8.3	35%	65%	3.61%	5.3%	–10.1%	–10.2%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Balanced, y 4	Stock-like	7.8	34%	66%	3.57%	5.1%	–9.5%	–9.7%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Cautious, y 8	Stable	7.7	18%	82%	3.18%	2.7%	–3.9%	–4.5%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								
Cautious, y 8	Cyclical	7.4	17%	83%	3.16%	2.6%	–3.5%	–4.1%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								
Cautious, y 8	Stock-like	6.9	15%	85%	3.10%	2.3%	–2.8%	–3.4%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								
AT 65, THE DAY THE PAYCHECKS STOP								
Aggressive, y 2	Stable	12.4	68%	32%	4.23%	10.2%	–21.9%	–20.6%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Aggressive, y 2	Cyclical	12.3	68%	32%	4.23%	10.2%	–21.9%	–20.6%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Aggressive, y 2	Stock-like	12.1	68%	32%	4.23%	10.2%	–21.9%	–20.6%
Savings dominate; the mix is close to the stand-alone optimum of 68% stocks.								
Balanced, y 4	Stable	10.9	34%	66%	3.58%	5.1%	–9.6%	–9.8%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Balanced, y 4	Cyclical	10.6	34%	66%	3.58%	5.1%	–9.6%	–9.8%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Balanced, y 4	Stock-like	9.9	34%	66%	3.58%	5.1%	–9.6%	–9.8%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 34%.								
Cautious, y 8	Stable	9.7	17%	83%	3.15%	2.6%	–3.5%	–4.0%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								

Risk	Pay	Savings, years of pay	Stocks	TIPS	Expected real return, compound	Volatility	Bad year, 1 in 100	Worst fall 2011-26
Cautious, γ 8	Cyclical	9.3	17%	83%	3.15%	2.6%	−3.5%	−4.0%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								
Cautious, γ 8	Stock-like	8.7	17%	83%	3.15%	2.6%	−3.5%	−4.0%
Little human capital is left. At a 3.1-point premium the stand-alone stock share is 17%.								

Each group follows its own expected path from 25 with median pay and a 12.1% saving rate, so savings differ a little across rows. The expected return is the compound rate of the mix; the bad year starts from its arithmetic mean. Volatility, the bad year and the worst fall count TIPS as held to maturity; marked to market as a 10-year TIPS fund, every row would show a fall of 26% to 30% since 2011, most of it in 2022. “Worst fall” is the largest real peak-to-trough loss of the mix from May 2011 to September 2026, rebalanced monthly. The bad year uses a Student-t with five degrees of freedom. My calculation.

Source: Model output; inputs from Table 1 and Table 2.

Four patterns are worth saying in words.

- **Down the ages, stable pay:** the stock share falls because k rises. A balanced saver goes from 100% at 30 to 66% at 40 and 34% at 65. The compound return she can expect falls from 4.61% to 3.58%, which is what it costs to stop being able to lose a third in a year.
- **Across risk aversion:** at any age past the early thirties the stock share is close to proportional to 1/γ. Doubling γ halves it.
- **Across paychecks:** the same person at 40 holds 66% in stocks with stable pay, 51% with cyclical pay and 28% with stock-like pay. By 65 the three are identical, because the paycheck is gone.
- **At the corners:** young savers with stable pay want more stock than they are allowed without borrowing, so they all hold 100% whatever their γ. Cautious savers with stock-like pay want less than zero, so they all hold none.

The calculator applies the same formula to any age, saving rate and amount saved.

The model for one investor

Set the inputs. The mix is the model's answer at November 2026 prices. It is an illustration, not advice for you.

Age	**45**	Retirement age	**65**
Risk aversion γ	**4.0 · Balanced**	Pay	**Stable (β = 0)**
Savings so far	**3.50 years of pay**	Savings rate	**12.1% of pay**



U.S. stocks **32%** International developed **14%** Emerging markets **5%** TIPS **49%**

Stocks
51%
of savings, at global market weights; the rest in TIPS

Expected real return
3.9%
a year, before fees and taxes

Volatility
7.7%
a year, with TIPS held to maturity

Bad year, 1 in 100
−15.8%
real return on the portfolio

Stocks trail TIPS
28%
chance over 20 years

Savings still to come
1.8×
years of pay, present value; 33% of retirement wealth

Savings at retirement
11.2×
final pay, if every year earned its expected return

Ladder income
55%
of final pay for 30 years, at today's real yields

Method: stock share = clip([π / (γσ²) − β(1 − k)] / k, 0, 1), with π = 3.09 points (the arithmetic premium: the compound premium plus half the variance), σ = 15.1%, k = savings so far ÷ (savings so far + present value of the savings still to come) and β the share of future pay that behaves like stocks. Pay follows the Census age profile [28], discounted on the real Treasury curve [1]. Retirees put 1 ÷ (funded ratio) of wealth in a TIPS ladder and invest the rest at π / (γσ²). "Stocks trail TIPS" uses a lognormal approximation. A stand-alone investor with this γ would hold 34% in stocks.

11. My own portfolio

A model that its author does not hold himself against is a lecture. So here is my portfolio at the November 9, 2026 close, in percentages only. View A is the liquid portfolio: a Roth IRA (37.5%), a 401(k) (15.2%), a taxable brokerage account (40.8%) and bank cash (6.5%). View B adds unvested restricted stock units in my employer's stock, which are about 35% of the total when included. I do not own those shares yet and cannot sell them, so the model target is measured against View A, and View B is how I look at risk.

Table 11. What I hold

Percent of the portfolio at the November 9, 2026 close. View A is the liquid portfolio; View B adds unvested employer stock

Holding	What it is	Account	View A	View B
SWTSX	U.S. total stock market index fund	Roth IRA	37.5%	24.4%
VTI	U.S. total stock market ETF	Taxable brokerage	22.8%	14.8%
TDF 2065	Target-date 2065 index fund	401(k)	15.2%	9.9%
VXUS	International stocks ex U.S., index ETF	Taxable brokerage	3.6%	2.3%
GGLL	2x daily leveraged ETF on one stock	Taxable brokerage	3.2%	2.1%
AVUV	U.S. small-cap value ETF	Taxable brokerage	2.5%	1.6%
SMH	Semiconductor sector ETF	Taxable brokerage	2.0%	1.3%
NOW	Single stock	Taxable brokerage	1.4%	0.91%
GOOGL	Single stock	Taxable brokerage	1.2%	0.78%
VT	Global stock index ETF	Taxable brokerage	1.1%	0.71%
SLV	Silver trust	Taxable brokerage	1.0%	0.65%
RVII	Listed closed-end fund of private companies	Taxable brokerage	0.65%	0.42%
Cash	Bank and brokerage cash	Bank 6.5%, brokerage 1.4%	7.9%	5.1%
Employer stock	Unvested restricted stock units	Not yet mine to sell	–	about 35%

The tickers are facts about my portfolio, not recommendations. The target-date fund's mix on September 30, 2026: U.S. stocks 58.68%, international developed 24.94%, emerging 7.99%, listed real estate 4.83%, bonds 2.91%, cash 0.65% [39]. RVII last traded at $15.85 on November 9, 2026 [40].

Source: My own records; fund holdings [39]; Nasdaq [40].

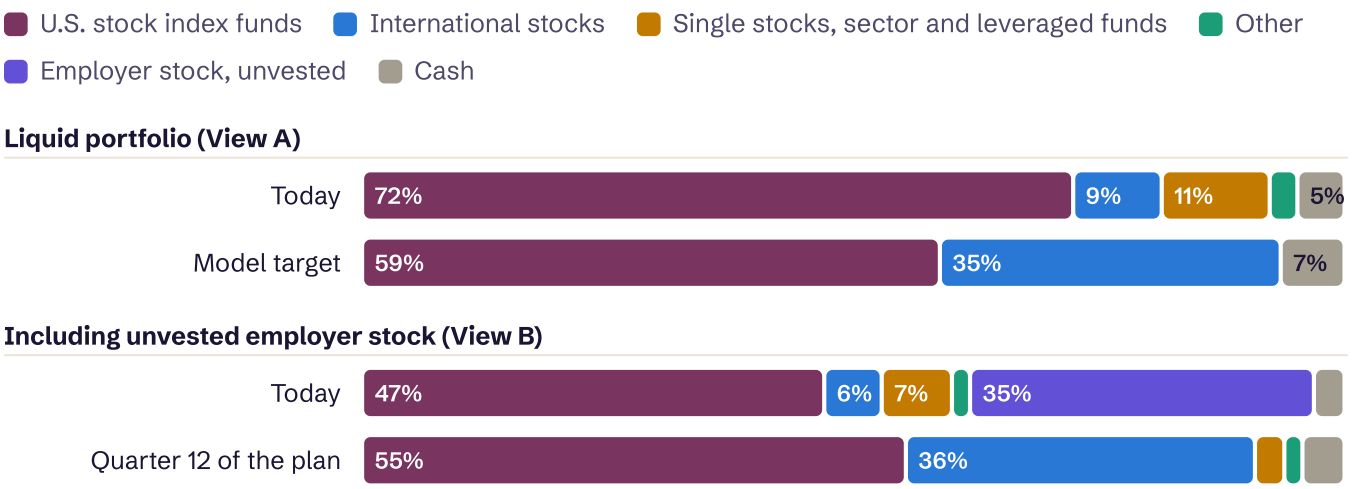
Where I sit in the model

I am in the first years of my career, which puts me in the youngest age band. My salary is stable, so β is 0 for View A. For risk aversion I asked myself what I would do if the portfolio fell 40% in a year, and the honest answer was buy more. That puts me at γ = 2, the most aggressive group on the grid. I do not go lower, because a γ of 2 already means 110% in stocks on the historical averages: below it the model describes someone who would have borrowed to buy stocks, not a saver.

For that group the formula sits at its corner. The stock budget is 67.9% of retirement wealth, and while savings so far are less than 68% of that wealth the portfolio is 100% global stocks. At the default saving rate that holds until savings pass about 6.7 years of pay; I treat my liquid portfolio as inside that range. I keep the bank cash as a reserve outside the model. The target for View A is therefore 58.8% U.S. stocks, 25.7% international developed, 9.0% emerging and 6.5% cash.

Figure 17. My portfolio against the model's target for my group

Percent of the portfolio by exposure, at the November 9, 2026 close; the target is measured against the liquid portfolio



The 2x fund counts at twice its weight with the borrowing netted against cash, so today's bars show exposure, not market value. "Other" is listed real estate and bonds inside the target-date fund, silver, and a listed fund of private companies. The last bar is the plan's end state, after the employer stock is sold in about quarter 8: it is not yet at target because the plan does not sell appreciated taxable positions.

Source: My own holdings; look-through of the target-date fund from its issuer-reported holdings [39]; global market weights [17].

Table 12. Today against the target, by exposure

Liquid portfolio (View A), percent

Exposure	Today	Target	Gap	After quarter 12
U.S. stock index funds	72.4%	58.8%	−13.6	55.4%
International developed stocks	6.8%	25.7%	+18.9	26.3%
Emerging-market stocks	2.3%	9.0%	+6.7	9.2%
Single stocks, sector and leveraged funds	11.0%	0.0%	−11.0	3.0%
Listed real estate (inside the target-date fund)	0.7%	0.0%	−0.7	0.5%
Bonds (inside the target-date fund)	0.4%	0.0%	−0.4	0.3%
Silver	1.0%	0.0%	−1.0	0.7%
Listed fund of private companies	0.7%	0.0%	−0.7	0.4%
Cash, net of the leveraged fund's borrowing	4.8%	6.5%	+1.7	4.3%

Stock exposure today is 93.8% of the liquid portfolio counting the 2x fund at twice its weight (90.6% at market value), and 89.6% of it is U.S. The target keeps 6.5% bank cash as a reserve outside the model and puts the rest in global stocks at market weights. The last column is larger than 100% of today's portfolio by the employer-stock proceeds; its weights are shares of the larger portfolio. My arithmetic.

Source: My own holdings; model output.

What the numbers say

- **The amount of risk is right. The mix is not.** My stock exposure is 93.8% of the liquid portfolio counting the leveraged fund at twice its weight, close to the model's 100% of the invested part. But 89.6% of it is U.S., against a global weight of 62.9% [17]. The target moves 25.7% of the portfolio abroad. On the model's inputs that raises expected real return from 3.95% to 4.42% and lowers volatility from 15.6% to 14.2% (my arithmetic).
- **About 10% of the portfolio is a deliberate bet.** A 2x daily leveraged fund on one stock (3.2%), two single stocks, a semiconductor fund and a small-cap value fund add up to 10.3% by weight. Counting the leveraged fund at twice its weight and leaving out the small-cap fund, which is diversified, single-name and sector exposure is 11.0%. Their risk apart from the market, measured on 2022–2026 monthly prices [40], adds 0.6 points to the portfolio's volatility. The model gives that risk no expected return, since it holds the market.
- **The leveraged fund has a cost that is not a matter of opinion.** A fund that resets to twice a stock's return every day compounds at twice the stock's rate less $(L^2 - L)/2 \cdot \sigma^2$ a year, which for $L = 2$ is σ^2. The stock's volatility since 2022 has been 31.7% [40], so the drag is about 10.1% a year (my arithmetic), before the fund's fees and financing.
- **Once it is counted, the employer stock is the main risk.** In View B the portfolio's volatility is 18.5%, and 27% of its variance is risk the market does not explain, tied to one company and its industry. I do not use my employer's price history. That figure uses a typical single stock: 30% volatility, with 83% of its variance coming from its industry and the firm itself, not the market [36]. Most individual stocks have returned less than Treasury bills

over their lives, and 4% of them account for all of the market's gain [37]. And this stock is tied to the same company as my paycheck, which makes my human capital more stock-like until the shares are sold.

- **Two small positions the model would not hold.** Silver has no yield. The listed fund of private companies is the illiquid sleeve of the previous section in an exchange-traded wrapper; it last traded at $15.85 on November 9, 2026 [40], and I have found no published net asset value since its listing to compare that with. Together they are 1.6%.

Pricing the concentration

How much extra return does a concentrated position need to earn its place? For an investor at the optimum, the last dollar in a position whose volatility apart from the market is σ must earn

$$\textit{required extra return} = \gamma \cdot k \cdot w \cdot \sigma^2$$

w: the position's share of the portfolio. k: savings as a share of total retirement wealth. γ: risk aversion. The more of your wealth is still future savings, the less a concentrated position in the portfolio matters.

Table 13. The return a concentrated position has to earn to pay for its own risk

Extra expected return a year, over a diversified holding with the same market exposure, at which the position is the right size for γ = 2

Savings as a share of total wealth (k)	Single names and sector, 11% of the liquid portfolio	Employer stock, about 35% of the total
10%	0.08%	0.52%
25%	0.19%	1.31%
50%	0.38%	2.61%
100%	0.77%	5.23%

Hurdle = γ × k × w × σ², with w the position's weight and σ the volatility the market does not explain: 18.7% for the basket of single names (measured on 2022–2026 monthly prices [40]) and 27.3% for a typical single stock, industry and firm risk together [36]. k is not published here; a saver at 25 with one year of pay in the bank has a k of 0.24 at the default saving rate. The hurdle leaves out the link between an employer's stock and the paycheck from the same employer, which makes the true number higher. My arithmetic.

Source: Model output; Campbell, Lettau, Malkiel and Xu [36]; Nasdaq prices [40].

Table 13 gives the hurdle for both. Early in a career k is small, so the single-name basket needs only a fraction of a percent a year to justify itself: 0.19% at k of one quarter. I am not going to pretend the model condemns it. The employer stock is different in size. At k of one quarter it needs 1.31% a year over the market and at one half 2.61%, and that leaves out its link to my salary. The model's verdict is to sell it when it can be sold. That is a statement about concentration, not about the company.

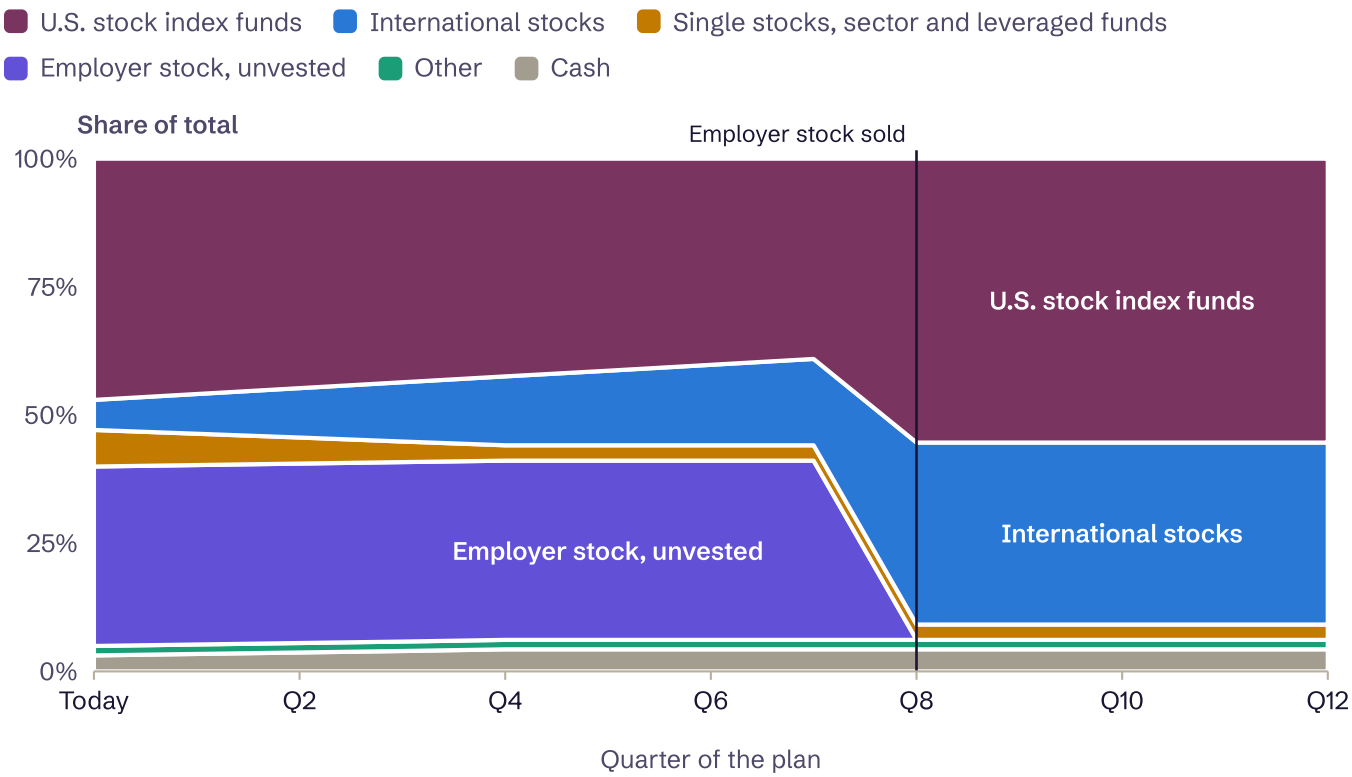
The plan: 12 quarters

I am not going to move in one day. The plan runs 12 quarterly steps, about three years, in the order that costs least in tax.

- **Exchanges inside the Roth IRA, every quarter.** The Roth holds one U.S. index fund. Moving part of it into international index funds realizes no tax. The international gap is 25.7% of the portfolio; cash and the leveraged fund's proceeds cover 4.6%, so the Roth has to supply 21.1%, or 1.76% of the portfolio per quarter. At that pace the liquid portfolio reaches its international weight in quarter 12 even if the employer stock is never sold.
- **The 401(k) stays where it is.** Its target-date fund is 96.4% stocks, 32.9 points of that international [39]. For my group that is already close to the target.
- **Idle brokerage cash and the leveraged fund, over the first four quarters.** Selling the leveraged fund is taxable, but the tax is paid once and the drag is paid every year. At a 15% rate on gains, which is my assumption, even if the whole position were gain the tax equals 1.5 years of drag; if half were gain, 0.7 years (my arithmetic). The proceeds go to international index funds, where the gap is.
- **New contributions go to whatever is furthest below target.** I have not published my contribution rate, so the base plan leaves them out and Table 14 shows what they would add.
- **Employer stock, once it is mine to sell. For planning I put that at about quarter 8.** It is the largest single source of new money, about 54% of today's liquid portfolio. Any sale happens only when the shares are mine and my employer's trading rules allow it. The proceeds go first to whatever is below target and the rest into U.S. index funds. That finishes the move abroad 4 quarters early, so the Roth exchanges stop at 14.1% moved instead of 21.1%.
- **Other taxable positions are left alone.** The single stocks, the sector fund, the small-cap value fund, silver and the private-company fund fail the model's test, but by Table 13 the cost of holding them is small at my k, and selling means paying tax now. They shrink as a share of a growing portfolio and are sold when there is a loss to harvest.

Figure 18. The plan: twelve quarterly steps, with the employer-stock sale at about quarter 8

Portfolio including unvested employer stock (View B), percent of the total at each quarter



No new contributions are included (I have not published a number); Table 14 shows what they would add. Prices are held flat, so every change in the picture is a trade. My arithmetic.

Source: My own holdings and the model target.

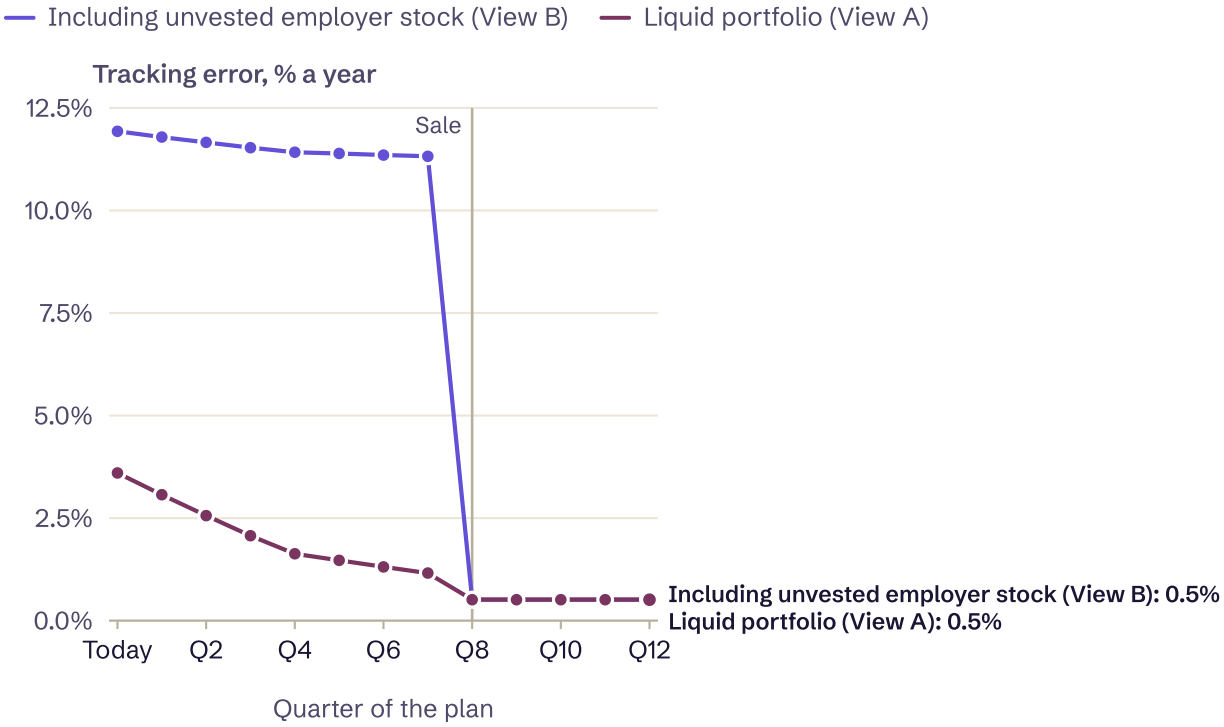
Figure 18 is the path with employer stock included. Figure 19 measures the distance to target each quarter. On the liquid portfolio the share that is off target falls from 27.4% to 14.1% in the first year, about 3.3 points a quarter, then 1.76 points a quarter, and to 4.9% once the employer stock is sold and reinvested. Tracking error on View A goes from 3.6% a year to 1.6% and then 0.5%. On View B it barely moves until the sale, from 11.9% to 11.3%, and then drops to 0.5%. Most of my distance from the model is one position I cannot trade yet.

$$TE = \sqrt{(d'\Sigma d + \Sigma_i w_i^2 \sigma_i^2)}$$

TE: tracking error against the target. d: portfolio weights minus target weights, by asset class. Σ: the model's covariance. w_i, σ_i: weight and volatility apart from the market of each single-name position.

Figure 19. Distance from the target, quarter by quarter

Tracking error against the model target, percent a year, for the liquid portfolio and with employer stock included



Tracking error is the volatility of the difference between the portfolio and the target: class weights through the 2022-2026 covariance, plus the risk the market does not explain in single names (from public prices [40]) and in a typical single stock for the employer shares (27.3% a year [36]). "Off target" is half the sum of absolute weight differences. My calculation.

Source: Model output.

Table 14. What new money would do to the path

A what-if, not my contribution rate: new money each quarter as a percent of today's liquid portfolio, all of it sent to whatever is furthest below target

New money a quarter	Off target after Q4	After Q8	After Q12	Tracking error after Q4	Single-name exposure at Q12
None (the base plan)	14.1%	4.9%	4.9%	1.63%	3.0%
1%	11.1%	4.7%	4.5%	1.34%	2.8%
2%	8.3%	4.4%	4.2%	1.09%	2.6%
3%	6.7%	4.2%	4.0%	0.88%	2.4%

Each 1% of the portfolio added per quarter takes about 0.7 points more off the gap each quarter in the first year, until international stocks reach target, then dilutes the legacy positions. Off target is half the sum of absolute weight differences on the liquid portfolio. My arithmetic.

Source: Model output.

What is left at the end is 4.9% off target: the legacy single names at 3.0%, silver, the private-company fund and the small positions inside the target-date fund. I can live with that. The plan does not depend on a forecast. If U.S. stocks keep winning for three more years I will have sold some of them too early, and I will still own 55% of the portfolio in them.

12. What would change my mind

The model is simple on purpose, and it is wrong in ways I can list.

- **The stock return.** Everything scales with the premium. At Damodaran's 6.51% the balanced retiree holds 55% in stocks, not 34%, and at the CAPE yield 11%. If real yields fell back to 1% with stock valuations unchanged, the compound premium would be 3.61 points and the share 52%. The model should be re-run when prices move; the scripts are in the repository for that reason.
- **Three conventions set the level.** I read yield-based returns as compound rates and add half the variance before the formulas use them. I name the risk groups off the old premium over bills. I use the ten-year TIPS yield as the safe rate at every age. Read the stock return as an arithmetic mean instead and the balanced share is 21%, not 34%. Name the groups off the old premium over Treasury bonds and "the 60/40 investor" holds 45%. Use the thirty-year TIPS yield and the balanced share is 29%. The direction is the same on every one. The level runs from about a fifth to under a half.
- **Future real yields.** The simulation invests every future contribution at today's real yields and prices the ladder at 65 on today's curve. If real yields fall back toward zero, a saver who leans on TIPS ends with less than Table 8 shows, and most of the model's edge over a target-date fund goes with it.
- **Returns are treated as independent from year to year.** If stocks mean-revert, they are less risky over long horizons than this model says and the young should hold more. If my return estimates are wrong in a persistent way, they are more risky and everyone should hold less. I have not modeled either.
- **β and γ are choices.** The three paycheck types and the six risk groups are labels I picked. The calculator lets a reader set their own.
- **What is left out.** Taxes, Social Security as an asset, housing, annuities, and investors outside the U.S. Social Security alone would raise most retirees' funded ratios and with them the stock share in Table 7.
- **The window is 57 months.** Table 5 shows the answer on longer data. If stocks and bonds go back to hedging each other, long Treasuries come back into the portfolio.

None of that changes the main result, which rests on two published numbers: a ten-year TIPS yield of 2.66% and stock valuations that imply about 4.61%. While those hold, the portfolio that most advice calls balanced holds 34% to 77% more stock than its owner would choose with the prices in front of them, and it holds the rest in the wrong bonds.

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